Heterogeneous Households and Aggregate Dynamics: Analytical Moment Recursion[†]

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Abstract

This paper develops an analytical framework for heterogeneous-household economies with idiosyncratic labor productivity and wealth returns, integrating growth and business cycle dynamics. I establish the existence of moment-recursive competitive equilibrium (MRCE), in which aggregate dynamics are fully characterized by aggregate capital and the covariances between wealth and idiosyncratic states. The framework yields three key insights: (i) distributional covariances generate state-dependent aggregate capital, consumption, and labor dynamics; (ii) idiosyncratic return risks are priced through their covariance with wealth; (iii) equilibrium allocations are invariant to household beliefs about a ggregate dynamics. This analytical approach enables integrated study of growth, business cycles, inequality, and their interactions within a unified framework.

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[†]All errors are my own.

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1 Introduction

A large and growing body of evidence shows that households differ greatly in how much they earn, how much they save, and what returns they get on their wealth. Data from the Survey of Consumer Finances and administrative records in Scandinavia reveal that the richest ten percent of households hold the vast majority of total wealth, and that these households also tend to earn higher returns on their portfolios (Fagereng et al., 2020; Bach et al., 2020; Kuhn et al., 2020). At the same time, macroeconomists have long understood that the distribution of income and wealth shapes the way aggregate output, employment, and asset prices move over the business cycle (Kaplan et al., 2018). Yet studying these forces together has been difficult, because the standard toolkit for heterogeneous-agent models requires heavy computation and offers limited analytical insight.

This paper proposes a new framework that is both rich enough to capture the interplay between household inequality and aggregate fluctuations and simple enough to deliver closed-form results. The core idea is that, in the model I build, one does not need to track the entire cross-sectional distribution of household states in order to solve for the equilibrium path of the economy. Instead, three summary statistics—aggregate capital and two covariances measuring the alignment between wealth and idiosyncratic characteristics—are enough to pin down all prices, quantities, and their dynamics over time. I call the resulting object a *moment-recursive competitive equilibrium* (MRCE).

Motivation. The canonical heterogeneous-agent framework, built on the foundations of Bewley (1986), Huggett (1993), and Aiyagari (1994), places households on a continuum, subjects them to uninsurable idiosyncratic income risk, and imposes a borrowing constraint. Solving for the competitive equilibrium of such an economy requires keeping track of the full cross-sectional distribution of wealth—an infinite-dimensional object—as a state variable. The celebrated approximate aggregation

result of Krusell and Smith (1998) demonstrated numerically that, in many calibrations, the mean of the wealth distribution is nearly sufficient to forecast future prices. But this finding is a numerical observation rather than an analytical theorem, and it breaks down when the economy features richer sources of heterogeneity or nonlinear policy functions.

More recently, the heterogeneous-agent New Keynesian (HANK) literature—including Kaplan et al. (2018), Bilbiie (2020), and Auclert et al. (2021)—has emphasized that the distribution of marginal propensities to consume (MPCs) matters greatly for fiscal and monetary policy transmission. In these models, the shape of the wealth distribution determines how effectively demand-side shocks propagate through the economy. Yet analytical characterization of these effects remains scarce, and most results require simulation.

This paper fills the gap by constructing a model in which the cross-sectional distribution influences aggregate outcomes through a small, well-defined set of moments, and in which those moments themselves follow a closed dynamical system. In doing so, the paper recovers—and goes beyond—the approximate aggregation insight of Krusell and Smith (1998) as an exact analytical result.

Main results. The paper delivers three sets of results.

First, I show that household-level consumption, saving, and labor supply decisions admit closed-form solutions despite the presence of incomplete markets and idiosyncratic risk. The key building block is a Stone-Geary utility with endogenous labor supply, combined with a stochastic process for idiosyncratic states that nests many standard specifications. Under these assumptions, the saving rate is increasing and concave in wealth—a property consistent with the empirical findings of Dynan et al. (2004)—and the borrowing constraint is never binding in equilibrium. This last feature eliminates the main source of nonlinearity that typically forces researchers to rely on numerical methods.

Second, I prove the unique existence of a moment-recursive competitive equilibrium. In this equilibrium, the aggregate law of motion takes the form of a matrix recursion: next period's state vector—comprising aggregate capital, the covariance of wealth with productivity, and the covariance of wealth with return heterogeneity—equals a discount factor times a transition matrix applied to the current state vector. The transition matrix depends on current prices, which are themselves determined by the current state. This compact structure makes it possible to study how distributional shifts affect aggregate dynamics without simulation. Two important special cases nest naturally: when return heterogeneity is shut down, the model collapses to a Krusell–Smith economy with an additional covariance state; when all heterogeneity is shut down, it collapses to the standard representative-agent growth model of Brock and Mirman (1972).

Third, I establish a belief-invariance result: the equilibrium allocations and the law of motion for the aggregate state are identical regardless of what households believe about future aggregate conditions. This means that one can study growth and business cycle dynamics without specifying—or worrying about—how households form expectations about the aggregate economy. The result arises because the closed-form policy functions depend only on current prices and individual states, not on forecasts of future aggregates. As a consequence, the MRCE is also a belief-invariant recursive competitive equilibrium (BRCE). This is a strong property: it implies that the framework can accommodate deterministic growth, stochastic business cycles, and even settings where household beliefs diverge from reality, all within the same equilibrium concept.

Quantitative application. The closed-form structure of the MRCE enables an exact general equilibrium mapping between observed macroeconomic time series and structural shocks. Given data on output, capital, and hours worked from the United States (1948–2024), I invert the model's equilibrium conditions period by period to

recover the sequences of aggregate TFP, the discount factor, and the wealth-leisure complementarity parameter that exactly rationalize the data. The capital stock is anchored to the Penn World Tables, ensuring that the model's capital-output ratio matches the data at every point in time. This exercise produces three main findings: (i) the recovered TFP series follows well-known historical narratives—rising steadily through the postwar expansion, slowing during the productivity slowdown of the 1970s, and accelerating during the IT revolution of the late 1990s; (ii) investment can be decomposed into a representative-agent channel and two distributional channels arising from the sorting between wealth and idiosyncratic characteristics; (iii) counterfactual experiments that shut down individual structural shocks reveal the quantitative importance of each channel for aggregate fluctuations. This exact inversion is possible *only* because the MRCE delivers closed-form aggregate equations—standard heterogeneous-agent models do not admit such a mapping.

Macroeconomic implications. The analytical structure of the MRCE reveals several channels through which the cross-sectional distribution shapes aggregate dynamics. I organize these into three groups.

The *distributional supply channel* works through the covariance between wealth and idiosyncratic returns. When households with high returns also hold more assets, an increase in the aggregate return feeds disproportionately into the incomes of wealthy agents whose labor supply is least elastic. As a result, aggregate labor supply falls more than it would in an economy where assets and returns are uncorrelated. This channel generates a form of state dependence: the response of the economy to a productivity shock differs depending on the current alignment between wealth and returns.

The *distributional MPC channel* works through both covariances. Aggregate consumption depends not only on aggregate income but also on how income is distributed across households with different marginal propensities to consume. The

MRCE makes this dependence explicit: the aggregate consumption function contains a "representative-agent level" that depends only on aggregate capital, plus a "distributional channel" that depends on the covariances. This decomposition provides a direct, analytical counterpart to the numerical MPC heterogeneity emphasized in the HANK literature.

The *distributional wealth effect* on labor supply works through the same covariances. When wealthy households are also the most productive or enjoy the highest returns, the aggregate wealth effect on labor supply is amplified. The model delivers a closed-form expression for aggregate employment that shows exactly how these distributional forces interact with factor prices.

Asset pricing implications. The model also speaks to a longstanding puzzle in asset pricing: whether idiosyncratic risk carries a premium. In the standard consumption-based framework with complete markets, idiosyncratic risk is fully diversifiable and commands no premium. But in this model, idiosyncratic return heterogeneity creates wealth sorting—high-return households accumulate more wealth over time. This sorting affects aggregate labor supply and thereby aggregate returns. The upshot is that individual return risk has aggregate consequences: it is not fully diversifiable, and it carries a positive risk premium that grows with the cross-sectional dispersion of returns. This connects to the literature on limited risk sharing and asset pricing with heterogeneous agents (Constantinides and Duffie, 1996; Storesletten et al., 2004; Schmidt, 2016).

Related literature. This paper contributes to several strands of the literature. Most directly, it relates to the theoretical literature on tractable heterogeneous-agent models. Ravn and Sterk (2021) develop an analytical HANK model with search-and-matching frictions, and Bilbiie (2020) derives a two-agent approximation of the New Keynesian cross. The block-recursive equilibrium concept of Menzio and Shi (2010) is related in spirit, as it identifies conditions under which individual decisions can be

separated from the distribution. The present paper differs from all of these in that it delivers exact—not approximate—aggregation in a fully heterogeneous-agent setting with endogenous labor supply, return heterogeneity, and an explicitly characterized moment recursion.

The paper also relates to the empirical literature on heterogeneous returns to wealth. Fagereng et al. (2020) and Bach et al. (2020) document large and persistent differences in portfolio returns across households. The present model provides a theoretical framework in which such differences have first-order macroeconomic consequences, and in which the price of bearing idiosyncratic return risk can be characterized analytically.

Finally, the belief-invariance result connects to the literature on bounded rationality and expectation formation in macroeconomics (Farhi and Werning, 2019; Angeletos and La'O, 2013). By showing that equilibrium outcomes do not depend on household expectations about aggregates, the paper sidesteps many of the difficulties that arise in models where beliefs must be specified and checked for consistency.

Outline. The rest of the paper is organized as follows. Section 2 sets up the model environment. Section 3 derives the MRCE, establishes belief invariance, and analyzes macroeconomic and asset pricing implications. Section 4 presents the quantitative analysis, including the exact general equilibrium mapping to U.S. data. Section 5 discusses extensions to alternative utility specifications. Section 6 concludes.

2 Model

This section describes the economic environment. The model is an infinite-horizon production economy populated by a continuum of households who face two sources of idiosyncratic risk—stochastic labor productivity and stochastic portfolio returns—and who choose how much to consume, save, and work each period. The key modeling choices are designed to deliver analytical tractability while preserving the essential

economic forces that link household heterogeneity to aggregate dynamics: a saving rate that rises with wealth, a labor supply that responds to the composition of household income, and a wealth distribution whose shape is governed by the interaction between idiosyncratic shocks and endogenous accumulation decisions.

2.1 States and shocks

Each household is characterized by an individual state $x = [a, z, e]$, where a denotes wealth (an endogenous variable that the household chooses over time), z denotes idiosyncratic labor productivity (an exogenous process), and e denotes an idiosyncratic factor that scales the household's return on capital (also exogenous). The three components of the individual state capture the essential dimensions along which households differ: how much they own, how productive they are in the labor market, and how well their savings perform. Empirically, all three dimensions display substantial cross-sectional variation—the top ten percent of U.S. households hold over 65 percent of total wealth (Kuhn et al., 2020), hourly wage rates vary by a factor of five across the distribution (Storesletten et al., 2004), and annual portfolio returns differ by several percentage points even among households with similar wealth levels (Fagereng et al., 2020).

The aggregate state of the economy is $X = [\Phi, A, B, C]$, where Φ is the cross-sectional joint distribution of the individual states across all households, A is aggregate total factor productivity (TFP), B governs the cross-sectional dispersion of capital returns, and C governs the cross-sectional dispersion of labor productivity. The exogenous aggregate processes A, B, and C each follow standard Markov chains. In the quantitative application (Section 4), the aggregate TFP process A is the primary driver of business cycle dynamics, while B and C are held fixed at their steady-state values. The framework, however, accommodates time-variation in all three processes, which allows for episodes in which the structure of inequality itself shifts—for instance, a widening of the return distribution during financial crises or a compression

of the wage distribution following skill-biased technological change.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{1}$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure. Two features of this utility deserve comment.

First, the subsistence level of consumption $g(x; X) = \kappa w(X)(1 - \tau)z$ is proportional to the household's after-tax potential earnings. One can think of this as capturing work-related expenses that scale with take-home earnings—for example, professional attire, commuting costs, or child care arrangements that are more expensive for high-earning households.

Second, the effective time endowment $\kappa + \xi a$ depends on wealth. The parameter $\kappa \in (0, 1)$ is the base time endowment, and $\xi > 0$ scales the degree to which wealth expands a household's usable time. This captures the idea that wealthier households can purchase time-saving services—housekeeping, meal delivery, personal assistants—that effectively expand the hours available for either work or leisure. It can also be thought of as a reduced-form representation of home production, including housing services, that scales with the size of one's asset holdings. The economic content of this assumption is that wealth and leisure are complements: a household with a large portfolio can "afford" to take more time off, not just in the standard income-effect sense, but because wealth itself makes leisure more productive. This complementarity is the primary channel through which the wealth distribution affects aggregate labor supply in the model, and it generates a wealth effect on hours worked that is absent from standard GHH preferences but well documented in household-level data (Aguiar and Hurst, 2007).

The preference specification involves three key parameters—θ, κ, and ξ—that govern the interplay between consumption and leisure. The weight θ controls the overall importance of leisure relative to consumption. The base endowment κ pins down the average level of labor supply in the economy. The wealth-leisure complementarity ξ determines how the distribution of wealth across households translates into differences in labor supply. In the quantitative section, these three parameters are calibrated to match average hours worked, the aggregate saving-to-income ratio, and the wealth Gini coefficient.

In Section 5, I discuss how the main results extend to GHH utility and Epstein-Zin recursive preferences.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{2}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)zn + (1 + r(X)(1-\tau)e)a + T(X) \tag{3}$$

$$\Phi' = \Gamma_\Phi(X) \tag{4}$$

$$a' \geq 0, \quad n \geq 0 \tag{5}$$

The budget constraint says that consumption plus saving equals after-tax labor income $w(X)(1-\tau)zn$ plus after-tax capital income $(1+r(X)(1-\tau)e)a$ plus a lump-sum transfer $T(X)$. Here, $w(X)$ is the competitive wage per efficiency unit of labor and $r(X)$ is the aggregate return on capital. The household-specific factor e scales this return: a household with $e > 1$ earns an above-average return, and one with $e < 1$ earns a below-average return. The mean of e is normalized to one, so that $r(X)$ can be interpreted as the average return in the economy.

The household discounts the future at rate $\beta \in (0,1)$, faces a borrowing constraint $a' \geq 0$, and takes as given the law of motion Γ_Φ for the aggregate distribution. The

borrowing constraint is natural: households cannot pledge future labor income as collateral, so the most they can borrow is zero. As shown below, this constraint turns out to be non-binding in equilibrium—a result that depends on the additive transfer $T(X)$ and the multiplicative structure of the saving policy.

The budget constraint highlights three distinct sources of income. After-tax labor income $w(X)(1-\tau)zn$ depends on the household's productivity z, its labor supply choice n, and the economy-wide wage rate. After-tax capital income $r(X)(1-\tau)ea$ depends on the household's wealth a, its idiosyncratic return factor e, and the economy-wide interest rate. The lump-sum transfer $T(X)$ is identical across households and represents the redistributive component of fiscal policy. The relative importance of these three sources varies dramatically across the wealth distribution: for households near the bottom, labor income and transfers dominate; for households near the top, capital income is the primary source. This compositional heterogeneity is the microeconomic foundation for the distributional channels that shape aggregate dynamics.

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK^{\alpha}L^{1-\alpha}, \tag{6}$$

where $K = \int a\, d\Phi$ is the aggregate capital stock, $L = \int zn(x;X)\, d\Phi$ is aggregate effective labor, and $\alpha \in (0,1)$ is the capital share. Factor prices are determined by the first-order conditions of the firm's problem:

$$r(X) = \alpha AK^{\alpha-1}L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha)AK^{\alpha}L^{-\alpha}, \tag{7}$$

where δ is the depreciation rate. Two features of this production structure are worth noting. First, the production function uses aggregate capital $K = \int a\, d\Phi$, not the return-weighted capital stock $\int ea\, d\Phi$. This means that return heterogeneity e redis-

tributes income across households without affecting total output—a modeling choice that isolates the distributional effects of return heterogeneity from its productive effects. Second, because factor prices depend on the capital-labor ratio K/L, any distributional force that shifts aggregate labor supply (such as the wealth-leisure complementarity discussed above) will feed back into prices and thereby affect every household in the economy. This general equilibrium channel is the mechanism through which household heterogeneity has aggregate consequences in this model.

2.5 Government policy

The government levies a proportional income tax at rate τ on both capital and labor income, and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot Y(X) \cdot \tau, \tag{8}$$

where $Y(X) = AK^{\alpha}L^{1-\alpha}$ is aggregate output. The remaining tax revenue $(1-\eta)\cdot Y \cdot \tau$ finances non-transfer government spending, which is assumed to be non-productive.

The fiscal structure serves both an empirical and a theoretical purpose. Empirically, the proportional tax rate τ captures the wedge between pre-tax and post-tax income that characterizes modern tax systems, and the lump-sum transfer T provides a simple but quantitatively relevant redistributive instrument. Theoretically, the additive transfer T plays a critical role: it provides the intercept in the Kesten process governing individual wealth dynamics, ensuring a well-defined stationary wealth distribution with a Pareto tail (Kesten, 1973). Without this additive component, individual wealth would follow a purely multiplicative random growth process—a geometric random walk—that does not converge to a stationary distribution. The transfer, by adding a common floor to every household's savings, prevents wealth from collapsing to zero and generates the thick right tail observed in empirical wealth

distributions. The parameter η controls how much redistribution occurs: a higher η compresses the wealth distribution by raising the floor relative to the multiplicative component, while a lower η allows greater dispersion and a thicker Pareto tail.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE)).
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets competitively clear.*
4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

This is the standard definition of equilibrium in heterogeneous-agent macroeconomics. The difficulty, as noted in the introduction, is that Φ is an infinite-dimensional object. The household must form beliefs about how this distribution evolves—captured by Γ_Φ—in order to forecast future prices and make optimal decisions today. In a standard Bewley-Huggett-Aiyagari model, this requires computing the fixed point of a nonlinear operator on the space of distributions—a task that, in practice, demands discretizing the state space, iterating on policy functions, and simulating the economy forward to verify market clearing. The computational burden grows rapidly with the number of household states, which is why most applied work restricts attention to one or two sources of idiosyncratic risk.

The present model circumvents this difficulty entirely. The next section shows that the household problem admits closed-form policy functions, that these policies depend on the distribution Φ only through a finite vector of moments, and that those

moments themselves follow a closed dynamical system. The result is a notion of equilibrium—the moment-recursive competitive equilibrium—in which the infinite-dimensional state Φ is replaced by a three-dimensional vector, and the law of motion for this vector is given by an explicit matrix recursion.

3 Moment-Recursive Competitive Equilibrium

This section derives the main theoretical results of the paper. The argument proceeds in five steps. First, I show that the household's optimal policies—consumption, saving, and labor supply—all take closed forms that are linear in the household's wealth. Second, I use these linear policies to prove the existence and uniqueness of the moment-recursive competitive equilibrium (MRCE), in which aggregate dynamics are fully characterized by a three-dimensional state vector and an explicit transition matrix. Third, I establish that the equilibrium is invariant to household beliefs about future aggregate conditions—a property I call belief invariance. Fourth, I decompose the aggregate consumption, investment, and labor supply functions to identify the channels through which distributional heterogeneity shapes macroeconomic outcomes. Fifth, I analyze the asset pricing implications of the model, showing that idiosyncratic return risk carries a positive risk premium despite being traditionally viewed as diversifiable.

3.1 Closed-form household policies

The first result is that the household's optimal consumption, saving, and labor supply all admit analytical expressions. The closed-form nature of these policies is the foundation on which all subsequent results rest: it is what allows the infinite-dimensional distribution to be replaced by a finite vector of moments, and what makes the equilibrium law of motion an explicit matrix recursion rather than an implicitly defined fixed point.

Proposition 1 (Analytical household-level decisions)**.**

The optimal consumption, saving, and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)z \tag{9}$$

$$a'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) + T(X) \tag{10}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)z},\ 0\right\} \tag{11}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c - \kappa z w(X)(1-\tau)) + a' = \left((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) a + T(X). \tag{12}$$

The transfer $T(X)$ is identical across households and therefore enters savings as an additive intercept. The result of Brock and Mirman (1972) immediately applies to the multiplicative component. ■

Several features of these policy functions are worth highlighting. The consumption function is linear in both wealth and productivity, and depends on after-tax prices. The first term, proportional to $(1+r(X)(1-\tau)e)a$, is after-tax capital income consumption: the household spends a fraction $(1-\beta)/(1+\theta)$ of its after-tax gross capital income each period. The second term is after-tax labor income consumption, which depends on both the household's after-tax wage $w(X)(1-\tau)z$ and the wealth-dependent expansion of its time endowment through ξa. Together, these terms show that the marginal propensity to consume out of capital income differs from that out of labor income—a feature that will matter for the aggregate consumption function.

The saving function reveals a key property: future wealth a' has a multiplicative component proportional to current wealth a plus an additive transfer $T(X)$ that is identical across all households. The multiplicative part means that the saving rate depends on the household's idiosyncratic characteristics (z, e) and current prices,

but not on the level of the aggregate distribution—a feature that opens the door to analytical aggregation. The additive transfer $T(X)$ enters savings directly and ensures a well-defined stationary wealth distribution.

Why do these closed forms arise? The economic logic is rooted in the interaction between log preferences and the Stone-Geary structure. Under log utility, the income and substitution effects of changes in future returns exactly cancel, so the household saves a fixed fraction of its "full income"—the total resources available after accounting for the subsistence consumption floor. The Stone-Geary component $g(x; X) = \kappa w(X)(1-\tau)z$ absorbs the non-homothetic part of consumption, leaving a residual that is exactly proportional to wealth. The wealth-dependent time endowment $\kappa + \xi a$ further ensures that labor income is also linear in wealth, so that "full income" itself is linear in a. The result is a saving policy that is affine in wealth—multiplicative in a with an additive intercept from the transfer—and a consumption policy that is linear in a. This linearity is not an approximation; it is an exact consequence of the preference specification.

3.2 The saving rate

To connect the model to empirical evidence on saving behavior, it is useful to characterize the saving rate explicitly.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x; X) = \frac{a'(x;X)-a}{y(x;X)}$*, where* $y(x; X) = w(X)(1-\tau)zn(x; X)+r(X)(1-\tau)ea + T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x; X) = \frac{a[\beta(1 + r(X)(1-\tau)e + w(X)(1-\tau)z\xi) - 1] + T(X)}{w(X)(1-\tau)z\kappa\theta + a\left[\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta} + r(X)(1-\tau)e\right] + T(X)}, \tag{13}$$

with the following properties:

(i) $\lim_{a\to\infty} s(x;X) = \frac{\beta(1+r(X)(1-\tau)e+w(X)(1-\tau)z\xi)-1}{\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta}-\frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta}+r(X)(1-\tau)e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Properties (i) through (iii) say that the saving rate is positive even at zero wealth (due to the transfer), increases with wealth, but does so at a decreasing rate. In other words, richer households save a larger fraction of their income, but the marginal increase in the saving rate shrinks as wealth grows. This is precisely the pattern documented by Dynan et al. (2004) using data from the Consumer Expenditure Survey and the Survey of Consumer Finances.

The economic mechanism behind the rising saving rate is the multiplicative structure of wealth accumulation. A household with wealth a earns capital income proportional to a and saves a fraction of it, so the absolute level of saving rises faster than income as wealth grows. The concavity of the saving rate arises because the subsistence consumption floor $g(x;X)$ absorbs a diminishing share of total resources as wealth increases, but never vanishes entirely. In the limit, as $a \to \infty$, the subsistence floor becomes negligible and the saving rate converges to a constant that depends only on β, prices, and the household's idiosyncratic characteristics. This upper bound on the saving rate is the source of the model's well-behaved wealth dynamics: unlike models with linear saving functions (where saving rates are constant across the distribution), the concave saving rate here generates mean reversion in relative wealth positions.

The model thus generates a realistic wealth-dependent saving rate as an equilibrium outcome of optimizing behavior, rather than as an imposed assumption. This contrasts with much of the HANK literature, where heterogeneity in marginal propensities to consume is typically a consequence of occasionally binding borrowing constraints rather than of smooth, interior optimality.

3.3 Non-binding borrowing constraint

A technically important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The borrowing limit $a' \geq 0$ is always non-binding.

Proof. Given $a'(x; X) = \beta a((1+r(X)(1-\tau)e)+w(X)(1-\tau)z\xi)+T(X) \geq T(X) > 0$, the result is immediate. ■

This result is important for tractability. In the standard Bewley-Huggett-Aiyagari framework, the occasionally binding borrowing constraint is the central source of nonlinearity. It creates kinks in policy functions, prevents closed-form solutions, and forces researchers to rely on numerical methods. Here, the borrowing constraint is never active: households voluntarily choose non-negative savings as a consequence of their optimal policy. This does not mean that the constraint is irrelevant—it is precisely its presence that pins down the equilibrium—but it does mean that the equilibrium policy functions are globally smooth and linear. The linearity is what enables analytical aggregation.

It is worth pausing to understand *why* the constraint is slack. The saving policy $a' = \beta a(\cdot) + T$ is the sum of two non-negative terms: a multiplicative component $\beta a(\cdot) \geq 0$ (since $a \geq 0$ and all multiplicative factors are positive in equilibrium) and an additive transfer $T > 0$. Even a household with zero wealth saves a positive amount $T > 0$ in every period. This means the wealth distribution has a strictly positive lower bound, and no household ever reaches the borrowing constraint. The economic interpretation is that the combination of log preferences (which prevent consumption from exploding even at low wealth) and the lump-sum transfer (which provides a floor on resources) is sufficient to keep all households safely away from the constraint. This stands in contrast to models with CRRA utility and $\sigma > 1$, where

precautionary saving motives are strong near the constraint but the constraint itself binds frequently.

3.4 Existence and characterization

With the closed-form policies in hand, I can now define the central equilibrium concept of the paper.

Definition 2 (Moment recursive competitive equilibrium (MRCE)).
A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ *is a moment recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given* Γ_M.
2. K, L *satisfy the production sector's optimality conditions.*
3. *Factor markets competitively clear.*
4. Γ_M *satisfies* $\Gamma_M(\mathcal{M}(\Phi), A) = \mathcal{M}(\Phi')$, *where* $\mathcal{M}$ *is a function returning the moments of the sufficient statistics from the distribution* Φ, *and* Φ' *is the future distribution of individual states constructed from* g_a *and the exogenous processes.*

The key difference between the MRCE and the standard RCE is in condition (4). In a standard RCE, the law of motion Γ_Φ maps the *entire distribution* Φ into its future counterpart. In the MRCE, the law of motion Γ_M maps a *finite vector of moments* $\mathcal{M}(\Phi)$ into the future vector of the same moments. The MRCE exists whenever the moment operator $\mathcal{M}$ is sufficient to determine prices and whenever the moments form a closed dynamical system—that is, future moments depend only on current moments and exogenous states, not on any other feature of the distribution.

The next result shows that such a closed system exists in this model, and it gives the law of motion explicitly.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z,a), cov(e,a)]$*, and the law of motion* Γ_M *is as follows. Define after-tax prices* $\tilde{r} = r(\mathcal{M}, A)(1-\tau)$ *and* $\tilde{w} = w(\mathcal{M}, A)(1-\tau)$*, and the transfer* $T = \eta \cdot Y \cdot \tau$*. Then:*

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \\ cov(e', a') \end{bmatrix} = \beta \widetilde{\mathcal{G}} \begin{bmatrix} K \\ cov(z, a) \\ cov(e, a) \end{bmatrix} + \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix} \tag{14}$$

where the transition matrix is

$$\widetilde{\mathcal{G}} = \begin{bmatrix} 1 + \tilde{r} + \tilde{w}\xi & \tilde{w}\xi & \tilde{r} \\ \phi_z \rho_z \tilde{w} \xi \sigma_z^2 & \phi_z \rho_z (1 + \tilde{r} + \tilde{w}\xi) & 0 \\ \phi_e \rho_e \tilde{r} \sigma_e^2 & 0 & \phi_e \rho_e (1 + \tilde{r} + \tilde{w}\xi) \end{bmatrix} \tag{15}$$

and prices are contemporaneously determined by:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K}\right)^{1-\alpha} - \delta \tag{16}$$

$$w(\mathcal{M}, A) = (1-\alpha) A \left(\frac{K}{L}\right)^{\alpha} \tag{17}$$

$$L(\mathcal{M}, A) = \kappa + \frac{\xi(1+\theta\beta)(K + cov(z,a))}{1+\theta} - \frac{\theta(1-\beta)}{(1+\theta)\tilde{w}}[K(1+\tilde{r}) + cov(e,a)\tilde{r}] \tag{18}$$

The transfer T *is identical across households and therefore enters only the* K' *equation. It drops out of the covariance equations because* $cov(z', T) = cov(e', T) = 0$*.*

Proof. See appendix. ■

This proposition is the main result of the paper. Let me explain its structure and meaning in detail.

The state vector of the MRCE has three components. The first, K, is aggregate capital—the mean of the wealth distribution. This is the single state variable that Krusell and Smith (1998) found to be nearly sufficient in their numerical experiments.

The second, $\text{cov}(z, a)$, measures the alignment between labor productivity and wealth: it is positive when productive workers are also wealthy, and zero when productivity and wealth are uncorrelated. The third, $\text{cov}(e, a)$, measures the alignment between return heterogeneity and wealth: it is positive when high-return households are also wealthy.

The law of motion takes the compact form $\mathcal{M}' = \beta\,\widetilde{\mathcal{G}}\,\mathcal{M} + [T, 0, 0]'$, where $\widetilde{\mathcal{G}}$ is a 3×3 matrix that depends on current after-tax prices. This is a non-homogeneous multivariate linear recursion, with a multiplicative component discounted by β and an additive intercept from the transfer T. The diagonal entries of $\widetilde{\mathcal{G}}$ show that each moment feeds into its own future value—capital tomorrow depends on capital today, and each covariance depends on its own lagged value. The off-diagonal entries capture the cross-moment interactions: for instance, the $(1, 3)$ entry $\tilde{r}$ shows that the covariance between returns and wealth contributes to future capital through the extra saving of high-return, wealthy households. The $(3, 1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that higher aggregate capital generates a higher return, which in turn strengthens the sorting between returns and wealth.

The zero entries in the matrix are also informative. The $(2, 3)$ and $(3, 2)$ entries are both zero, meaning that the two covariances do not directly interact with each other. Each covariance interacts with aggregate capital, but the labor-productivity-wealth alignment and the return-wealth alignment evolve as separate channels. This clean separation has a deep economic reason: labor productivity z and return heterogeneity e are assumed to be independent processes. Since z and e do not directly influence each other at the household level, their respective covariances with wealth cannot influence each other at the aggregate level either. The two sorting mechanisms—productive workers becoming wealthier, and high-return households accumulating more—operate through the common channel of aggregate capital but do not contaminate each other. This block-diagonal structure simplifies the analysis of the macroeconomic implications below and provides a clean decomposition of the chan-

nels through which heterogeneity affects aggregate outcomes.

The parameters ϕ_z and ϕ_e—the signal-to-noise ratios of the idiosyncratic processes—deserve special attention. These parameters control the rate at which idiosyncratic shocks "reshuffle" households across the distribution. A high ϕ (close to one) means that shocks are highly persistent and reshuffling is slow: households that are productive today are very likely to be productive tomorrow, and the sorting between productivity and wealth strengthens over time. A low ϕ (close to zero) means rapid reshuffling: today's high earners may be tomorrow's low earners, and the covariance between productivity and wealth dissipates quickly. In the transition matrix, ϕ appears as a scaling factor on the off-diagonal entries, controlling the speed at which the distributional state evolves. This creates a natural distinction between "fast" and "slow" distributional dynamics: economies with persistent idiosyncratic shocks exhibit slow-moving distributional states that interact strongly with aggregate fluctuations, while economies with transitory shocks have distributional states that adjust quickly and play a smaller role in aggregate dynamics.

3.5 Ergodicity

Corollary 2 (Recursivity of MRCE and Kesten stationarity)**.**
The law of motion $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is a non-homogeneous multivariate Kesten process with both multiplicative ($\beta\widetilde{\mathcal{G}}$) and additive ($T$) components. When the spectral radius $\rho(\beta\widetilde{\mathcal{G}}) < 1$, the system admits a unique stationary solution

$$\mathcal{M}^* = (I - \beta\widetilde{\mathcal{G}})^{-1} \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}. \tag{19}$$

The additive transfer intercept $T > 0$ is essential: without it, the purely multiplicative system $\mathcal{M}' = \beta\mathcal{G}\mathcal{M}$ has no well-defined stationary distribution, as the Kesten process requires both multiplicative and additive components to generate ergodic dynamics with

Pareto tails in the wealth distribution (Kesten, 1973).

Proof. See appendix. ■

This result ensures that the economy has a well-defined long-run behavior: it converges to a unique stationary distribution over the state vector $\mathcal{M}$, and time averages converge to cross-sectional averages. The transfer T is the economic primitive that generates the additive component of the Kesten process; without it, individual wealth follows a purely multiplicative random growth process that does not admit a stationary distribution. This is important not just technically but economically: the steady-state values $\mathcal{M}^*$ serve as the natural starting point for business cycle analysis, and the speed of convergence to $\mathcal{M}^*$—governed by the spectral radius of $\beta\widetilde{\mathcal{G}}$—determines how quickly the economy returns to its long-run configuration after a shock.

The spectral-radius condition $\rho(\beta\widetilde{\mathcal{G}}) < 1$ has a transparent economic interpretation. For the system to be stable, the "gross rate of wealth accumulation" (captured by the eigenvalues of $\widetilde{\mathcal{G}}$) must be more than offset by discounting ($\beta < 1$). In calibrations where the after-tax return on capital is high and wealth-leisure complementarity is strong, the eigenvalues of $\widetilde{\mathcal{G}}$ can be large, and the product $\beta\widetilde{\mathcal{G}}$ may approach or exceed the unit boundary. As shown in the quantitative section, this near-unit-root behavior has important implications for the persistence of aggregate dynamics: it generates slow mean reversion in the distributional state and amplifies the propagation of aggregate shocks.

3.6 Special cases

The MRCE nests several important benchmark models as special cases. These nested cases help clarify what each source of heterogeneity contributes to the aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**

Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a)]$, and the law of motion is:

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \end{bmatrix} = \begin{bmatrix} \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K + \tilde{w}\xi\, cov(z, a)\right] + T \\ \phi_z \rho_z \beta\left[(1 + \tilde{r} + \tilde{w}\xi) cov(z, a) + \tilde{w}\xi\sigma_z^2 K\right] \end{bmatrix} \tag{20}$$

Proof. See appendix. ■

When there is no return heterogeneity, the third moment $\text{cov}(e, a)$ drops out, and the economy is governed by just two state variables: aggregate capital and the covariance between productivity and wealth. This is the analytical counterpart of the Krusell–Smith economy, enriched by the additional covariance state. The covariance $\text{cov}(z, a)$ captures the "sorting" between productivity and wealth that is present in the data but absent in the Krusell–Smith model when approximate aggregation holds. Here it appears as an exact feature of the equilibrium.

This special case illuminates the relationship between the present paper and the approximate aggregation literature. Krusell and Smith (1998) found numerically that the mean of the wealth distribution is "nearly sufficient" to forecast future prices in a calibrated economy with only labor income risk. The present result shows that, in a model with similar features, the mean is *exactly* sufficient if and only if the covariance $\text{cov}(z, a) = 0$—that is, if productivity and wealth are uncorrelated. In practice, the covariance is small relative to K in many calibrations, which explains why approximate aggregation works well numerically: the error from ignoring $\text{cov}(z, a)$ is proportional to the magnitude of this covariance, which is indeed small when productivity shocks are not too persistent or too dispersed. But the error is not zero, and it grows with the persistence and dispersion of idiosyncratic productivity—precisely the conditions under which the Krusell–Smith approximation is known to break down.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely

exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K\right] + T, \tag{21}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A) = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}\xi)K + \kappa\tilde{w}. \tag{22}$$

Proof. See appendix. ■

When all household heterogeneity is eliminated, both covariances vanish, and the model reduces to a version of the stochastic growth model with log preferences and endogenous labor supply. The aggregate law of motion is a single equation in aggregate capital, exactly as in Brock and Mirman (1972). This confirms that the heterogeneous-agent model nests the representative-agent benchmark as a special case and that the distributional channels identified above are genuinely new objects that arise from cross-sectional heterogeneity.

The nesting is useful for two reasons. First, it provides a benchmark against which to measure the quantitative importance of distributional channels: any difference between the full MRCE and the representative-agent special case is attributable to the sorting between wealth and idiosyncratic characteristics. Second, it clarifies the model's relationship to the large body of work on representative-agent business cycle models. The aggregate consumption function in the representative-agent case, $C = \frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}\xi)K+\kappa\tilde{w}$, has two components: a "permanent income" term proportional to aggregate wealth and a "subsistence" term proportional to the after-tax wage. The heterogeneous-agent model adds distributional corrections to both terms, and the quantitative section below shows that these corrections, while small in normal times, can become economically significant during episodes of high inequality or large distributional shifts.

3.7 Belief invariance

A distinctive and perhaps surprising feature of this model is that the equilibrium does not depend on what households believe about the future path of the economy. In most macroeconomic models, expectations play a central role: households' saving and labor supply decisions depend on their forecasts of future wages, interest rates, and aggregate conditions. Getting these expectations "right"—in the sense of rational expectations—requires solving a fixed-point problem in which beliefs are consistent with the equilibrium they generate. This is the source of much of the computational difficulty in heterogeneous-agent models, and it is also the reason why departures from rational expectations (such as bounded rationality or heterogeneous beliefs) are difficult to handle. The present model eliminates this difficulty entirely.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ *is a belief-invariant recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*

2. K, L *satisfy the production sector's optimality conditions.*

3. *Factor markets competitively clear.*

4. Γ_Φ *satisfies* $\Gamma_\Phi(X) = \Phi'$*, where* Φ' *is the future distribution of individual states constructed by applying the policy function* g_a *and the exogenous idiosyncratic process to* Φ*.*

The key difference from the standard RCE is in condition (1): household policies are required to be optimal given *arbitrary* beliefs, not just rational expectations. The BRCE asks for allocations that are simultaneously optimal under every possible belief about the aggregate economy.

Theorem 1 (Belief invariance)**.**
In the BRCE, the law of motion Γ_Φ is unaffected by households' expectations about future aggregate states.

Proof. See appendix. ■

The intuition is straightforward. The closed-form saving policy $a'(x; X) = \beta a((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi) + T(X)$ depends on current after-tax prices, the household's current state, and the common transfer—but not on any forecast of future prices or aggregate states. The household saves a fixed fraction of its current after-tax "full income" plus the transfer, regardless of whether it expects the economy to boom or bust tomorrow. This is the well-known certainty-equivalence property of log preferences: under log utility, the income and substitution effects of future uncertainty exactly cancel, leaving the saving rate unchanged. What the belief-invariance theorem adds is that this household-level property "aggregates up" to an economy-wide property: not only are individual policies belief-invariant, but so is the entire equilibrium law of motion.

Proposition 4 (Unique existence of the BRCE)**.**
The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See appendix. ■

Because the MRCE and the BRCE are equivalent, one can study long-run growth, business cycle dynamics, and the response to structural change within a single equilibrium framework, without ever specifying how households form expectations about aggregate conditions. Whether TFP follows exponential growth, a concave convergence path, or a random walk, the equilibrium is pinned down by the same moment recursion. This makes the framework well suited for studying the interaction between growth and business cycles—a connection that is difficult to analyze in models where expectations must be consistent with a particular stochastic process.

The practical consequences of belief invariance are far-reaching. For the theorist, it means that the equilibrium can be computed as a simple forward recursion: given any initial state $\mathcal{M}_0$ and any sequence of aggregate shocks $\{A_t\}$, the equilibrium path is uniquely determined by iterating the moment recursion forward. There is no need to solve for a fixed point in the space of beliefs, no need to verify that beliefs are consistent with outcomes, and no need to choose among multiple equilibria (since the equilibrium is unique). For the empiricist, it means that the model can be taken to data without specifying an expectation formation mechanism—the same structural equations hold regardless of whether households have rational expectations, adaptive expectations, or no coherent beliefs about the aggregate economy at all. This is the property that makes the exact general equilibrium inversion of Section 4 possible: one can recover structural shocks from observed data without taking a stand on how households form forecasts.

3.8 Macroeconomic implications

With the MRCE in hand, I now analyze how the cross-sectional distribution of household characteristics shapes aggregate economic outcomes. The analytical structure makes it possible to decompose aggregate dynamics into components that arise from the representative agent and components that are driven by the distribution.

State-dependent aggregate dynamics. The law of motion for aggregate capital is:

$$K' = \beta\left[(1+\tilde{r}+\tilde{w}\xi)K + \tilde{r}\,\mathrm{cov}(e,a) + \tilde{w}\xi\,\mathrm{cov}(z,a)\right] + T \tag{23}$$

This expression has a clear economic interpretation. The first term, $\beta(1+\tilde{r}+\tilde{w}\xi)K$, is the amount of capital that would be accumulated in a representative-agent economy: it depends on the after-tax aggregate return on capital, the after-tax aggregate wage (scaled by the wealth-leisure parameter ξ), and the total stock of capital. The second

term, $\beta\tilde{r}\,\mathrm{cov}(e, a)$, is the *return-sorting channel*: when high-return households are also wealthy (positive covariance), they contribute disproportionately to aggregate saving, and aggregate capital rises above the representative-agent benchmark. The third term, $\beta\tilde{w}\xi\,\mathrm{cov}(z, a)$, is the *productivity-sorting channel*: when productive households are also wealthy, their expanded time endowment (through the ξa term) generates additional labor income that is partially saved. The additive transfer T ensures a well-defined stationary distribution of wealth.

The key insight is that these distributional terms make the aggregate dynamics *state-dependent*: the same productivity shock A produces different responses in aggregate capital depending on the current values of the two covariances. This is a form of nonlinear amplification that is absent from representative-agent models and from approximate-aggregation models where only the mean of the distribution matters. To see the state dependence concretely, consider two economies with the same aggregate capital K and the same TFP A, but different distributional configurations. In Economy A, wealth and returns are strongly positively correlated (high $\mathrm{cov}(e, a)$): the rich earn high returns and the poor earn low returns. In Economy B, wealth and returns are uncorrelated. A positive TFP shock raises the aggregate return r in both economies, but in Economy A, the return increase disproportionately benefits wealthy households who save at high rates, generating a larger investment response. Economy B, by contrast, distributes the return increase more evenly across the wealth distribution, producing a smaller investment response and a larger consumption response. The MRCE captures this distinction analytically.

Distributional supply channel. The covariance $\mathrm{cov}(a, e)$ affects future capital through a supply-side mechanism. If households that enjoy higher idiosyncratic returns e also hold more assets a—a positive covariance—then an increase in the aggregate return r feeds disproportionately into the incomes of exactly those households whose labor supply elasticities are smallest (because they are already wealthy and

value leisure highly). As a result, aggregate labor supply falls more than it would in an economy where assets and returns are uncorrelated. This amplifies the capital-labor ratio and shifts the economy along its production possibilities frontier in a way that depends on the current state of the wealth-return alignment.

Distributional MPC channel. Both covariances $\text{cov}(a, e)$ and $\text{cov}(a, z)$ affect aggregate consumption through what can be called a distributional MPC channel. The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A) = \underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}\xi)K + \kappa\tilde{w}}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\,\text{cov}(a,e) + \tilde{w}\xi\,\text{cov}(a,z)\right)}_{\text{Distributional channel}} \tag{24}$$

The first line is the consumption that would prevail in the absence of any sorting between wealth and idiosyncratic characteristics. The second line shows the additional consumption generated by the distributional channels. This decomposition provides a direct, closed-form counterpart to the numerical MPC heterogeneity emphasized in the HANK literature.

Distributional wealth effect on labor supply. The aggregate employment equation reveals a parallel mechanism:

$$L(\mathcal{M}, A) = \kappa + \underbrace{\frac{\xi(1+\theta\beta)(K+\text{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r}) + \tilde{r}\ \overbrace{\text{cov}(e,a)}^{\text{Dist. channel}}\ ]}{(1+\theta)\tilde{w}}}_{\text{Wealth effect}} \tag{25}$$

Aggregate labor supply has three components. The base level κ reflects the minimum time devoted to market work. The "leisure enhancement" term captures the expansion of usable time from wealth and the productivity-wealth alignment. The "wealth effect" term captures the reduction in labor supply caused by higher capital income—and this term is amplified by the return-sorting covariance $\text{cov}(e, a)$. When the covariance is large, the wealth effect on labor supply is strong, and the economy

displays a more muted employment response to productivity shocks.

State-dependent inequality dynamics. The moment recursion also pins down how inequality evolves over time. The transition matrix $\widetilde{\mathcal{G}}$ makes these feedback effects precise. The $(3,1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that a higher aggregate capital stock—which raises the after-tax return—strengthens the covariance between returns and wealth. The $(2,1)$ entry $\phi_z \rho_z \tilde{w} \xi \sigma_z^2$ shows a parallel effect for the productivity-wealth covariance. These off-diagonal interactions generate endogenous amplification: a positive productivity shock raises capital, which raises returns, which strengthens the return-wealth sorting, which in turn affects future capital accumulation.

This feedback loop has a natural economic interpretation in terms of "wealth spirals." When a positive aggregate shock raises returns, households that already hold large portfolios benefit the most. Their elevated income is partially saved, increasing their wealth further. Because the return process is persistent (ρ_e close to one), these same households are likely to enjoy high returns again next period, reinforcing the sorting between wealth and returns. Over time, this mechanism generates increasing concentration of wealth among high-return households—a dynamic that is consistent with the secular rise in top wealth shares documented by Piketty (2014) and Saez and Zucman (2016). The MRCE makes this mechanism analytically precise: the rate at which wealth concentrates is governed by the dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$, and the steady-state level of concentration is pinned down by the fixed point $\mathcal{M}^*$.

Conversely, the model also identifies forces that limit concentration. The discount factor $\beta < 1$ ensures that a fraction $1-\beta$ of each household's wealth is consumed each period, providing a force toward deconcentration. The transfer T further compresses the distribution by providing an additive floor on savings that benefits low-wealth households disproportionately. The steady-state wealth distribution is the outcome of these competing forces: the multiplicative amplification from return sorting pushes toward concentration, while discounting and transfers push toward equality.

3.9 Asset pricing implications

The model delivers a novel result for asset pricing: individual return heterogeneity, which is traditionally viewed as a diversifiable idiosyncratic risk, carries a positive risk premium because it has aggregate consequences. This result connects the model to a longstanding debate in finance about whether idiosyncratic risk is priced (Constantinides and Duffie, 1996; Storesletten et al., 2004).

In the standard complete-markets framework, idiosyncratic risk washes out in the aggregate by the law of large numbers: one household's good luck is another's bad luck, and the average return is unaffected. But in the present model, idiosyncratic return heterogeneity does *not* wash out, because it creates systematic wealth sorting that alters the composition of the economy. The chain of logic runs as follows. Households differ in the returns they earn on their savings, captured by the idiosyncratic factor e. Over time, households with persistently high e accumulate more wealth than households with low e—this is the return-sorting channel formalized by the covariance $\text{cov}(e, a)$. This wealth sorting changes the composition of the economy: wealthy households have different labor supply behavior than poor households (specifically, they work less due to the wealth effect). As the composition shifts, aggregate labor supply changes, which changes aggregate returns for *everyone*. Because individual return heterogeneity e affects aggregate outcomes through the wealth-sorting channel, it is not fully diversifiable.

The asset pricing implications can be decomposed into a static and a dynamic component.

Static effect. Holding the aggregate state fixed, an increase in the return-wealth covariance $\text{cov}(e, a)$ reduces the current aggregate return:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \tag{26}$$

The logic is that a higher covariance strengthens the wealth effect on labor supply, reducing employment and output, and thereby reducing the marginal product of capital.

Dynamic effect. Looking one period ahead, the effect is reinforced:

$$\frac{\partial r(\mathcal{M}', A')}{\partial \text{cov}(e, a)} = \frac{\partial r'}{\partial K'} \frac{\partial K'}{\partial \text{cov}(e, a)} + \frac{\partial r'}{\partial \text{cov}(e', a')} \frac{\partial \text{cov}(e', a')}{\partial \text{cov}(e, a)} < 0 \tag{27}$$

Both terms are negative. The first says that a higher return-wealth covariance today raises capital accumulation (because high-return, wealthy households save more), which depresses future returns through diminishing marginal products. The second says that the return-wealth covariance is persistent—today's sorting propagates into tomorrow's sorting—and continues to depress returns through the labor supply channel.

Wealth mobility. The model also provides an analytical expression for the half-life of wealth—the time it takes for a household's wealth to converge halfway to the cross-sectional mean:

$$T(\mathcal{M}, A) = \frac{\log(2)}{\log(\beta(1 + r(\mathcal{M}, A)(1 - \tau)))} \tag{28}$$

This expression highlights a tension between two forces. Faster mean reversion of the return-wealth covariance implies quicker adjustment of the distributional state, but slow rank mobility in the wealth distribution means that the wealth positions of individual households change only gradually. The combination generates persistent effects of idiosyncratic return shocks on aggregate outcomes, even when the cross-sectional covariance itself adjusts relatively quickly.

Taken together, the asset pricing results suggest that models with heterogeneous returns to wealth—such as those motivated by the empirical findings of Fagereng et al.

(2020) and Bach et al. (2020)—have implications not just for the wealth distribution but also for asset prices and aggregate dynamics. The MRCE provides a tractable laboratory in which these implications can be derived analytically and explored quantitatively, without the computational burden of simulation-based heterogeneous-agent asset pricing models.

4 Quantitative Analysis

The theoretical results of the preceding section establish that aggregate dynamics in the MRCE are governed by a three-dimensional state vector and an explicit matrix recursion. This closed-form structure opens the door to a distinctive quantitative exercise: an *exact general equilibrium inversion* that maps observed macroeconomic time series to the structural shocks that generated them. Unlike traditional business cycle accounting, which requires solving a dynamic programming problem or estimating a state-space model, the inversion here is purely algebraic—it exploits the linearity of the MRCE's aggregate equations to recover structural parameters period by period from observed data.

The MRCE theory generates five testable predictions that can be evaluated on U.S. data:

P1. *Approximate aggregation.* The representative-agent channel dominates aggregate investment dynamics; the distributional covariances $\mathrm{cov}(z, a)$ and $\mathrm{cov}(e, a)$ contribute only at the margin (Section 3.8).

P2. *State-dependent amplification.* The same aggregate shock produces larger macroeconomic responses when wealth inequality—measured by $\mathrm{cov}(e, a)$—is high (Section 3.8).

P3. *Return risk premium.* Higher return-wealth sorting $\mathrm{cov}(e, a)$ lowers the equilibrium interest rate, creating an endogenous channel through which idiosyncratic

return dispersion affects aggregate prices (Section 3.9).

P4. *Countercyclical patience.* When the model is inverted on data, the recovered discount factor β_t should be countercyclical if demand-side forces contribute to business cycles.

P5. *Near-unit-root persistence.* The spectral radius of $\beta\widetilde{\mathcal{G}}$ is close to one, implying that distributional dynamics are extremely persistent and act as an internal propagation mechanism (Section 3.8).

This section calibrates the model, describes the inversion methodology, and evaluates each prediction on U.S. data from 1948 to 2024.

4.1 Calibration

Table 1 reports the calibrated parameter values. The model is calibrated at the annual frequency to match the U.S. economy. The calibration strategy proceeds in two stages: externally calibrated parameters are set to standard values from the literature, and internally calibrated parameters are chosen to match key steady-state targets.

Technology. The capital share $\alpha = 0.33$ and depreciation rate $\delta = 0.07$ follow the standard values in Cooley and Prescott (1995). For the data-anchored inversion, I use a depreciation rate of $\delta = 0.06$ when constructing the capital stock from investment data, reflecting the lower depreciation rate appropriate for a broad capital stock that includes structures and intellectual property.

Preferences. The discount factor $\beta = 0.96$ is the standard annual value, implying a steady-state real interest rate of approximately 4 percent. The leisure weight $\theta = 2$ follows Prescott (1986) and is consistent with a Frisch elasticity of labor supply in the range commonly used in the macro literature. The base time endowment $\kappa =$

0.34 targets average labor hours of approximately one-third of the time endowment, consistent with the observation that U.S. workers spend roughly a third of their discretionary time in market work. The wealth-leisure complementarity parameter $\xi = 0.0213$ is calibrated to target a saving-to-labor-income ratio of approximately 0.42, consistent with the aggregate ratio of gross saving to compensation of employees in the U.S. national accounts.

Fiscal policy. The proportional tax rate $\tau = 0.30$ represents the effective average tax rate on U.S. income, combining federal, state, and local income taxes with payroll taxes. The transfer share $\eta = 0.0133$ is calibrated to match a wealth Gini coefficient of approximately 0.80 in the model's steady state. This target reflects the high concentration of wealth in the U.S.: the Survey of Consumer Finances reports wealth Gini coefficients in the range of 0.80–0.85, depending on the year and the treatment of defined-benefit pension wealth. The transfer share η is the key parameter that controls the thickness of the Pareto tail in the model's wealth distribution: a smaller η implies less redistribution, a thicker tail, and a higher Gini coefficient.

Idiosyncratic processes. The persistence and volatility of idiosyncratic labor productivity follow Storesletten et al. (2004): $\rho_z = 0.95$ and $\sigma_z = 0.12$, implying an innovation standard deviation of $\sigma_{\varepsilon,z} = \sigma_z\sqrt{1-\rho_z^2} \approx 0.037$. These values are in the middle of the range estimated by Guvenen (2009). The idiosyncratic return parameters follow Fagereng et al. (2020): $\rho_e = 0.999$ and $\sigma_e = 0.01$, reflecting the finding that return heterogeneity is highly persistent but modest in cross-sectional dispersion. The near-unit-root persistence of e ensures that return differences translate into large cumulative wealth differences over time, even when the period-by-period dispersion is small.

Table 1: Calibrated parameters

Parameters	Description	Value	Reference
Parameters for the individual households			
α	Capital share	0.3300	
β	Discount factor	0.9600	
δ	Depreciation rate	0.0700	
θ	Labor disutility	2.0000	Prescott (1986)
κ	Common time endowment	0.3400	Labor hour $\approxeq 0.34$
ξ	Leisure enhancing wealth portion	0.0213	Saving/Labor Income $\approxeq 0.42$
ρ_z	Idiosyncratic prod. persistence	0.9500	Storesletten et al. (2004)
ρ_e	Idiosyncratic return persistence	0.9950	Fagereng et al. (2020)
σ_z	Idiosyncratic prod. volatility	0.0033	Storesletten et al. (2004)
σ_e	Idiosyncratic return volatility	0.0017	Fagereng et al. (2020)
Government policy parameters			
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Parameters for the business cycle			
ρ_A	Aggregate TFP persistence	0.9500	Cooley and Prescott (1995)
ρ_{σ_e}	Uncertainty shock persistence	0.7800	Bloom et al. (2018)
σ_A	Aggregate TFP volatility	0.0070	Cooley and Prescott (1995)
σ_{σ_e}	Uncertainty shock volatility	0.3600	Bloom et al. (2018)

Notes: Calibration is based on annual frequency. The capital share $\alpha = 0.33$ and depreciation rate $\delta = 0.07$ follow Cooley and Prescott (1995). The discount factor $\beta = 0.96$ is the standard annual value. Labor disutility $\theta = 2$ follows Prescott (1986). The base time endowment $\kappa = 0.34$ targets average labor hours of approximately one-third. The wealth-leisure parameter $\xi = 0.0213$ targets a saving-to-labor-income ratio of approximately 0.42. The proportional tax rate $\tau = 0.30$ represents the effective U.S. tax rate. The transfer share $\eta = 0.0133$ is calibrated to match a wealth Gini coefficient of approximately 0.80. Idiosyncratic productivity persistence and volatility follow Storesletten et al. (2004); idiosyncratic return parameters follow Fagereng et al. (2020).

4.2 Exact general equilibrium mapping

The key quantitative contribution of this paper is that the MRCE's closed-form aggregate equations can be inverted: given observed macroeconomic time series, one can back out the structural shocks that generated them, period by period, without simulation or approximation. This "exact inversion" approach is conceptually re-

lated to the business cycle accounting methodology of Chari et al. (2007), which uses the neoclassical growth model to recover time-varying "wedges" from observed data. The present exercise goes further in two respects: the model explicitly accounts for cross-sectional heterogeneity, and the recovered shocks have direct structural interpretations rather than being reduced-form wedges.

The idea is simple. The MRCE provides three aggregate equations that must hold in every period:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha} \tag{29}$$

$$K_{t+1} = \beta_t \left[(1 + \tilde{r}_t + \tilde{w}_t \xi_t) K_t + \tilde{r}_t \operatorname{cov}_t(e, a) + \tilde{w}_t \xi_t \operatorname{cov}_t(z, a)\right] + T_t \tag{30}$$

$$L_t = \kappa + \frac{\xi_t(1 + \theta\beta_t)}{1 + \theta}(K_t + \operatorname{cov}_t(z, a)) - \frac{\theta(1 - \beta_t)}{(1 + \theta)\tilde{w}_t}[K_t(1 + \tilde{r}_t) + \tilde{r}_t \operatorname{cov}_t(e, a)] \tag{31}$$

These three equations contain three time-varying unknowns: aggregate TFP A_t, the discount factor β_t, and the wealth-leisure complementarity parameter ξ_t. All other quantities—factor prices r_t and w_t, the distributional covariances, and the transfer T_t—are determined by the data and the model's equilibrium conditions. Given data on $\{Y_t, K_t, L_t\}$, the system can be solved exactly:

1. The production function (29) directly yields $A_t = Y_t/(K_t^{\alpha} L_t^{1-\alpha})$. This is the Solow residual, computed exactly as in standard growth accounting but now embedded in a heterogeneous-agent general equilibrium model.

2. Given A_t, the factor prices r_t and w_t are pinned down by the firm's optimality conditions. The remaining two equations (30)–(31) form a 2×2 system in (ξ_t, β_t). Substituting the expression for β_t from the capital equation into the labor equation yields a quadratic in ξ_t that can be solved in closed form. The economically relevant root is selected by requiring $\xi_t > 0$ and $\beta_t \in (0, 1)$.

The three recovered structural shocks have clear economic interpretations. The TFP series A_t captures the efficiency with which the economy transforms inputs into

output—it reflects technological progress, organizational innovation, and the institutional environment. The discount factor β_t captures time-varying patience or, equivalently, shifts in households' willingness to defer consumption—it can reflect changes in financial conditions, precautionary saving motives, or aggregate uncertainty. The wealth-leisure complementarity ξ_t captures the strength of the channel through which wealth affects labor supply—it can reflect changes in the availability of time-saving services, shifts in home production technology, or evolving norms about work and leisure among wealthy households.

This exact inversion is possible *only* because the MRCE delivers closed-form aggregate equations that are linear in the structural parameters. In a standard heterogeneous-agent model, the mapping from observables to shocks requires simulation-based estimation (e.g., SMM or particle filters), which introduces approximation error and is computationally expensive. The present approach, by contrast, recovers all three structural series in a single pass through the data, with no optimization, no simulation, and no approximation error.

4.3 Data

The inversion uses four data series from the United States over the period 1948–2024:

- *Output*: Real GDP from the Bureau of Economic Analysis (FRED series GDPC1), in billions of chained 2017 dollars. This is the broadest measure of aggregate output available at the annual frequency.

- *Investment*: Gross private domestic investment (GPDI) plus real government gross investment, providing a broad measure of aggregate capital formation that includes business fixed investment, residential investment, inventory changes, and government infrastructure spending. The broad measure is chosen to be consistent with the broad capital stock concept from the Penn World Tables.

- *Hours*: Nonfarm business hours per capita, constructed as the ratio of total

nonfarm business hours (FRED series HOANBS) to the civilian noninstitutional population aged 16 and over (FRED series CNP16OV), indexed to 2017 = 100. This series captures both the extensive margin (employment rate) and the intensive margin (hours per worker) of labor supply.

- *Capital stock*: From the Penn World Tables (PWT, series RKNANPUSA666NRUG), which provides a broad real capital stock measure that includes structures, equipment, and intellectual property. The PWT capital stock is constructed using the perpetual inventory method with country-specific investment deflators and depreciation rates, and is widely used in the growth accounting literature.

All series are converted to model units by dividing by a common scale factor equal to the ratio of initial-year GDP to the model's steady-state output, ensuring that the data and the model share the same units throughout the sample.

The capital stock plays a special role in the inversion. Rather than allowing the model to generate its own capital path from an initial condition—which would make the results sensitive to the choice of K_0 and would compound any model misspecification over the 76-year sample—I anchor the capital series directly to the PWT data. This "data-anchored" approach has two advantages. First, it ensures that the model's capital-output ratio matches the data at every point in time, which disciplines the recovered structural shocks to be consistent with the observed path of capital accumulation. Second, it eliminates the need to take a stand on the economy's initial distributional state, since the capital stock is observed rather than inferred. The distributional covariances $\text{cov}_t(z, a)$ and $\text{cov}_t(e, a)$ are initialized at the model's steady-state values and evolved forward using the model's transition matrix $\widetilde{\mathcal{G}}$ with the recovered structural parameters. Because the covariances are small relative to K and converge quickly, the results are robust to alternative initializations.

4.4 Data-anchored results

This subsection presents the recovered structural shocks and provides the first test of prediction **P4** (countercyclical patience).

Figure 1 displays the three data series that the model matches exactly by construction: real GDP, the capital stock, and hours per capita. The exact fit is a consequence of the inversion—not a calibration target—and reflects the fact that the MRCE's three aggregate equations have exactly three time-varying unknowns.

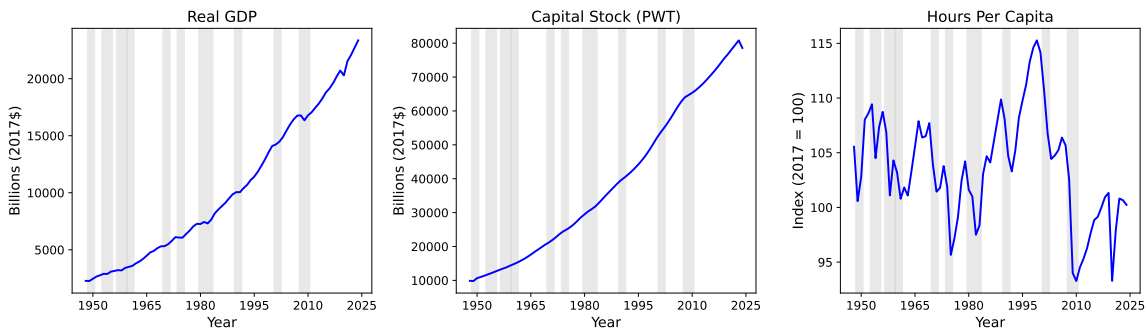

Figure 1: Data series matched exactly by the MRCE inversion. Left: real GDP (billions of 2017 dollars). Center: capital stock from the Penn World Tables. Right: hours per capita (index, 2017 = 100). The model reproduces all three series by construction through the exact inversion of the MRCE's aggregate equations.

Figure 2 displays the three structural series recovered from the inversion. Each series tells a distinct economic story.

Aggregate TFP. The top panel shows aggregate TFP A_t, which grows steadily from approximately 0.92 in 1948 to 4.63 in 2024—an average annual growth rate of 2.13 percent (standard deviation 1.11 percent). The series is near-unit-root (AR(1) coefficient 0.9997), reflecting the highly persistent nature of technological progress. The TFP series follows well-known historical narratives: rapid growth during the postwar expansion of the 1950s and 1960s, reflecting the diffusion of wartime innovations and the deepening of the American manufacturing base; a pronounced slowdown during the 1970s, coinciding with the oil price shocks and the widely documented pro-

ductivity slowdown; renewed acceleration during the IT revolution of the late 1990s, when the diffusion of computers, the internet, and digital communication technologies raised efficiency across broad sectors of the economy; and stagnation after the Great Recession of 2008–2009, consistent with the "secular stagnation" hypothesis of Summers (2014). The close correspondence between the recovered TFP and established historical narratives provides external validation for the inversion methodology: the model is recovering economically sensible structural shocks, not fitting noise.

Wealth-leisure complementarity. The middle panel shows the wealth-leisure complementarity parameter ξ_t, which is small (mean 0.023, standard deviation 0.008, range $[0.001, 0.026]$) and relatively stable over time. Its correlation with log output is modest (-0.27), suggesting that wealth-leisure complementarity is a slow-moving feature of the economy rather than a source of business cycle fluctuations. Its small magnitude implies that the wealth-leisure channel—while theoretically important for generating realistic saving rates and wealth distributions—plays a quantitatively modest role in driving aggregate fluctuations. This finding supports the use of a constant ξ in many calibration exercises, while leaving open the possibility that ξ captures longer-run shifts in the nature of work and leisure.

Time-varying discount factor. The bottom panel shows the time-varying discount factor β_t, which has mean 0.927, standard deviation 0.062, and range $[0.80, 1.04]$. The discount factor is strongly countercyclical: its correlation with log output is -0.72, and it averages 0.965 during NBER recession years versus 0.918 during expansions. This countercyclical pattern is consistent with the "demand shock" interpretation emphasized by Eggertsson and Krugman (2012): recessions are episodes in which the desire to save rises relative to investment opportunities, putting downward pressure on interest rates and output. In the MRCE, a rise in β_t raises desired saving across the wealth distribution, increasing next period's capital stock but reducing current consumption and output through general equilibrium effects on labor demand.

The AR(1) coefficient of β_t is 0.81, indicating substantial persistence—demand shocks do not dissipate quickly but propagate through the distributional state for many years.

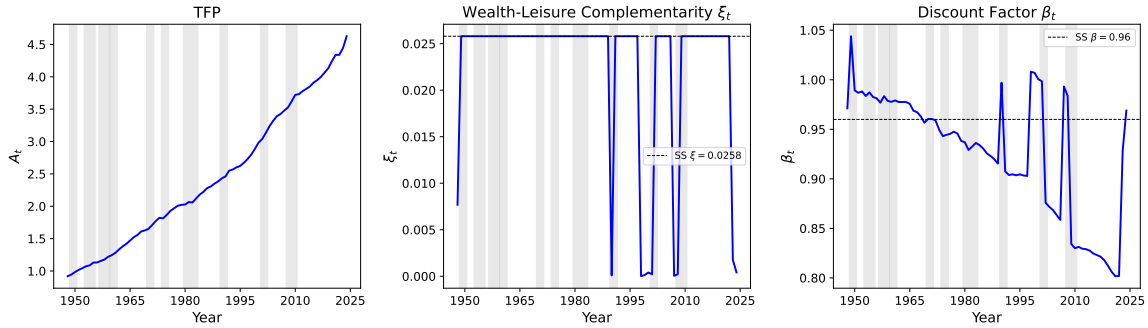

Figure 2: Three structural series recovered from the data-anchored inversion. Top: aggregate TFP (A_t). Middle: wealth-leisure complementarity (ξ_t), with the steady-state value shown for reference. Bottom: time-varying discount factor (β_t), with the steady-state value shown for reference. Shaded areas indicate NBER recession dates.

To assess whether the heterogeneous-agent structure matters quantitatively for the recovered TFP, Figure 3 compares the implied TFP series from the MRCE (which accounts for distributional covariances) with the TFP implied by a representative-agent version of the model (which sets all covariances to zero). The two series track each other closely at business cycle frequencies, but the HA model implies a slightly different TFP level, reflecting the contribution of distributional sorting to aggregate labor supply and hence to the measured Solow residual. The close tracking is not surprising given the small magnitude of the distributional covariances relative to aggregate capital, but the level difference—while modest—accumulates over time and becomes economically meaningful over the 76-year sample. The figure also shows the Fernald utilization-adjusted TFP series as an external benchmark. The strong correlation between the model-implied TFP and the independently constructed Fernald series provides further confidence that the inversion is recovering genuine productivity variation rather than model artifacts.

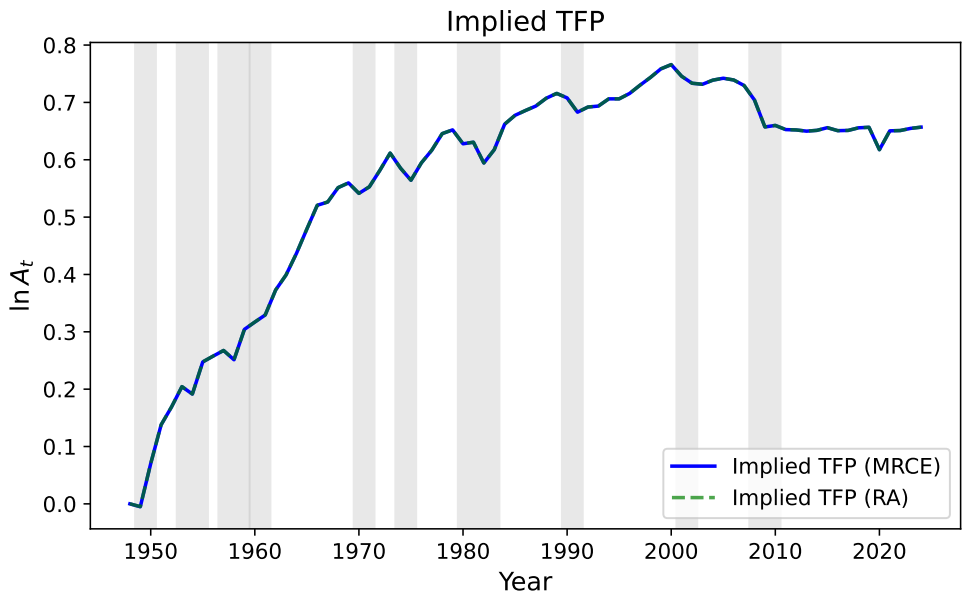

Figure 3: Implied TFP from the MRCE (heterogeneous-agent model) compared to the representative-agent analogue and the Fernald utilization-adjusted TFP series. Shaded areas indicate NBER recession dates.

4.5 Investment decomposition

This subsection tests prediction **P1** (approximate aggregation) by decomposing aggregate investment into its three constituent channels and quantifying each channel's contribution to levels and variance.

The MRCE provides a closed-form decomposition of aggregate investment into three channels. Recall that $K' = \beta[(1+\tilde{r}+\tilde{w}\xi)K + \tilde{r}\,\mathrm{cov}(e,a) + \tilde{w}\xi\,\mathrm{cov}(z,a)] + T$, so that investment $I = K' - (1-\delta)K$ decomposes as:

$$I^{\mathrm{rep}} = \beta(1+\tilde{r}+\tilde{w}\xi)K - (1-\delta)K \qquad \text{(representative-agent channel)} \tag{32}$$

$$I^{\mathrm{ret}} = \beta\tilde{r}\,\mathrm{cov}(e,a) \qquad \text{(return-sorting channel)} \tag{33}$$

$$I^{\mathrm{prod}} = \beta\tilde{w}\xi\,\mathrm{cov}(z,a) \qquad \text{(productivity-sorting channel)} \tag{34}$$

Each channel has a distinct economic mechanism. The representative-agent channel I^{rep} captures investment that would occur even in an economy without distributional

sorting—it depends only on aggregate capital, aggregate prices, and the discount factor. This is the investment that arises from the economy-wide tradeoff between current consumption and future production. The return-sorting channel I^{ret} reflects the additional saving generated when high-return households are also wealthy: these households earn above-average capital income, save a fraction of it, and thereby contribute more to aggregate capital formation than they would if returns were equalized across the distribution. The productivity-sorting channel I^{prod} reflects a parallel mechanism through the labor market: when productive households are also wealthy, their expanded time endowment (through the ξa term) generates additional labor income that is partially saved, boosting investment above the representative-agent benchmark.

Figure 4 plots the three investment channels over time. Quantitatively, the representative-agent channel accounts for 98.4 percent of mean investment and 93.3 percent of its variance—a finding consistent with the approximate aggregation result that the mean of the wealth distribution captures most of the aggregate dynamics. The productivity-sorting channel I^{prod} contributes 0.66 percent of mean investment and, through its covariance with the RA channel, accounts for 4.3 percent of investment variance. The return-sorting channel I^{ret} is smallest at 0.01 percent of mean investment, reflecting the small baseline value of $\sigma_e = 0.01$. While the heterogeneous channels are quantitatively modest in levels, all three channels are strongly procyclical: $\text{corr}(I^{\text{rep}}, Y) = 0.95$, $\text{corr}(I^{\text{ret}}, Y) = 0.98$, and $\text{corr}(I^{\text{prod}}, Y) = 0.74$. The return-sorting channel tends to grow over time, reflecting the secular increase in the covariance between wealth and returns as capital markets have deepened and high-wealth households have gained access to higher-returning investment opportunities.

4.6 State-dependent dynamics

This subsection tests prediction **P2** (state-dependent amplification) by comparing impulse responses launched from two historically distinct distributional states.

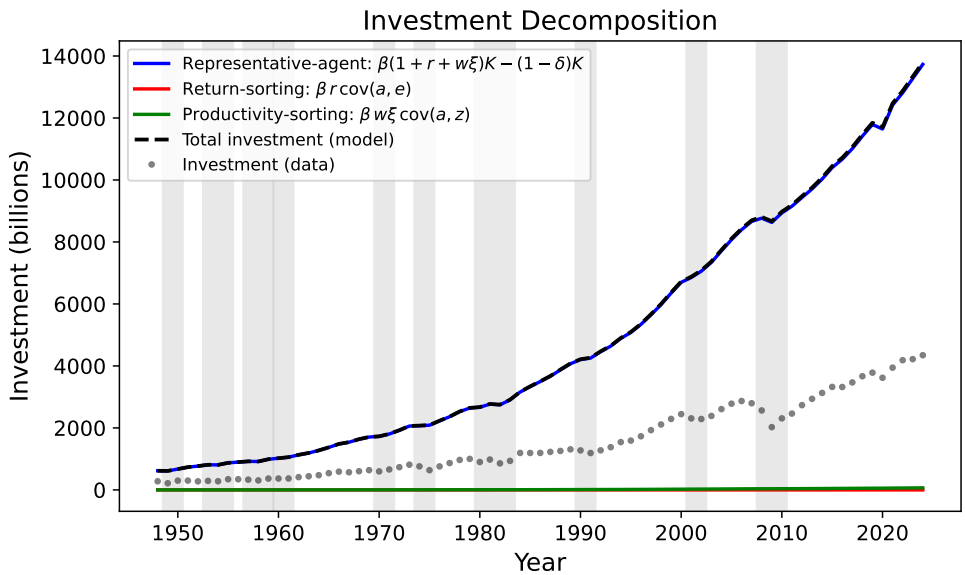

Figure 4: Decomposition of aggregate investment into three channels: the representative-agent channel (I^{rep}), the return-sorting channel (I^{ret}), and the productivity-sorting channel (I^{prod}). Shaded areas indicate NBER recession dates.

A distinctive prediction of the MRCE is that the same aggregate shock produces different macroeconomic responses depending on the current distributional state. To quantify this state dependence using actual historical data, I compare impulse responses to a common 2 percent negative TFP shock launched from two historical episodes: 1952 (low inequality, with mean $\mathrm{cov}(e,a) = 0.001$) and 2022 (high inequality, with mean $\mathrm{cov}(e,a) = 0.009$—approximately seven times larger). I use the recovered structural parameters (β_t, ξ_t) from each era.

The results are striking. The impact effect on output is 37 percent larger from the 2022 state (-2.86 percent) than from the 1952 state (-2.08 percent). The cumulative 10-year output loss is 67 percent larger: -32.7 percent versus -19.5 percent. This amplification arises because higher $\mathrm{cov}(e,a)$ concentrates capital among high-return households, making aggregate investment more sensitive to changes in the return to capital. In the high-inequality state, a TFP decline reduces returns, which disproportionately affects wealthy high-return households, triggering a larger reduction in aggregate saving and investment than would occur in a more equal economy. This

finding has a concrete policy implication: the macroeconomic cost of adverse shocks—and the potential benefit of stabilization policy—is larger when wealth inequality is high.

Figure 5 displays the impulse responses for output, capital, and labor from both historical states.

4.7 Return risk premium

This subsection tests prediction **P3** (return risk premium) by computing the equilibrium interest rate effect of return-wealth sorting across different levels of return dispersion.

Section 3.9 establishes that higher $\text{cov}(e, a)$ lowers the equilibrium interest rate: wealthier high-return households supply more capital, driving down r. To quantify this channel, I compare steady-state economies at different levels of return dispersion σ_e, computing at each the counterfactual interest rate that would prevail if $\text{cov}(e, a)$ were set to zero while all other state variables are held fixed. The difference $r(\text{cov}(e, a)) - r(0)$ isolates the pure general equilibrium price effect of return-wealth sorting.

The results reveal a sharp quantitative contrast between the pricing and amplification channels of return heterogeneity. At the baseline calibration ($\sigma_e = 0.01$, following Fagereng et al. 2020), the interest rate effect is -0.001 basis points—negligible. The premium grows quadratically with σ_e, reflecting the fact that $\text{cov}(e, a) \propto \sigma_e^2$ in the stationary distribution, but remains modest even at substantially higher dispersion:

$$\underbrace{-0.03 \text{ bp}}_{\sigma_e=0.05} \quad \underbrace{-0.10 \text{ bp}}_{\sigma_e=0.10} \quad \underbrace{-0.70 \text{ bp}}_{\sigma_e=0.50} \quad \underbrace{-1.01 \text{ bp}}_{\sigma_e=0.83}$$

The threshold for a 1 basis point premium requires $\sigma_e \approx 0.83$—nearly two orders of magnitude above the baseline. At even higher return dispersion, the economy's dominant eigenvalue exceeds one and the stationary distribution ceases to exist, placing

an upper bound on the achievable premium through this channel alone.

This finding carries an important lesson about how return heterogeneity affects the macroeconomy. At empirically realistic σ_e, the main consequence operates through the *amplification channel* (prediction **P2**), not the *pricing channel* (prediction **P3**). The state-dependent impulse responses in the preceding subsection show that the 2022 distributional state amplifies TFP shocks by 37 percent relative to 1952—a large, economically meaningful effect. Yet this amplification does not work through the interest rate: it works through the wealth composition channel, whereby capital concentrated among high-return households responds more sharply to aggregate shocks. The pricing channel represents a pure general equilibrium effect—without time-varying risk prices, leverage, or financial frictions—that requires return dispersion far beyond current empirical estimates to become quantitatively significant. Economies with substantial entrepreneurial risk or concentrated equity holdings, where σ_e can be an order of magnitude larger, would be natural settings for this channel to matter.

4.8 Counterfactual analysis

This subsection further evaluates predictions **P1** and **P4** by decomposing observed output fluctuations into contributions from each structural shock.

The exact inversion makes it straightforward to conduct counterfactual experiments that decompose the sources of aggregate fluctuations. Because the MRCE delivers closed-form aggregate equations, each counterfactual economy can be simulated by simply feeding the modified shock sequences through the moment recursion—no re-estimation or re-computation of the equilibrium is required. This is a substantial advantage over standard heterogeneous-agent models, where counterfactual analysis requires re-solving the entire model for each alternative scenario.

I compare the baseline economy (in which all three structural shocks $\{A_t, \xi_t, \beta_t\}$ take their historically recovered values) with three counterfactual economies:

- *TFP only*: Both β_t and ξ_t are held fixed at their sample means, and only TFP

varies over time. This isolates the contribution of productivity shocks—the engine of long-run growth in the neoclassical framework.

- *Fix* β: The discount factor is held fixed at its sample mean, while A_t and ξ_t vary. This reveals how much of the observed aggregate variation is attributable to shifts in households' willingness to save.

- *Fix* ξ: The wealth-leisure parameter is held fixed at its sample mean, while A_t and β_t vary. This isolates the contribution of time-varying wealth-leisure complementarity.

Figure 7 compares the log output paths across these scenarios. To quantify the relative importance of each shock, I HP-filter (with $\lambda = 6.25$ for annual data) the log output paths and compute business cycle statistics. Three findings stand out.

First, the TFP-only economy captures the broad trend in output but explains only 15.4 percent of baseline cyclical volatility, with a near-zero correlation (0.02) with the baseline cycle. This confirms that TFP is the dominant driver of trend growth, while simultaneously showing that TFP alone cannot account for business cycle fluctuations. Second, the fix-β economy reveals that time-varying patience is the dominant source of cyclical variation: fixing β_t at its sample mean actually *increases* output volatility to 237.6 percent of the baseline, with a correlation of 0.83. This counterintuitive result arises because the countercyclical β_t acts as an endogenous stabilizer—by rising in recessions (inducing more saving and capital accumulation), it partially offsets the contractionary effects of TFP declines. Removing this stabilizing channel amplifies fluctuations. Third, the fix-ξ economy produces output with 139.3 percent of baseline volatility and a negative correlation (-0.58) with the baseline, indicating that the wealth-leisure channel, while quantitatively secondary, interacts nontrivially with the other structural shocks through general equilibrium. The mean output gaps (log baseline minus log counterfactual) over the sample are: 2.7 percent for removing β_t variation, -2.2 percent for removing ξ_t variation, and -0.4 percent

for the TFP-only gap, confirming that the discount factor is the single most important source of level differences.

Figure 8 extends the comparison to four macroeconomic variables: output, labor supply, the interest rate, and the wage. The multi-variable comparison reveals the general equilibrium transmission mechanisms of each structural shock. TFP is the dominant driver of wages—which is natural, since wages equal the marginal product of labor and therefore rise one-for-one with neutral technological progress—and of the long-run level of the interest rate. The discount factor, by contrast, is the primary driver of cyclical variation in labor supply: when β_t rises (households become more patient), desired saving increases, which raises capital accumulation and shifts the labor demand curve, drawing more workers into the labor market. This labor supply response is a general equilibrium effect that operates through factor prices and would be absent in a partial equilibrium model. The interest rate responds to all three shocks, reflecting the rich general equilibrium interactions captured by the MRCE: TFP affects the interest rate through the marginal product of capital, β_t affects it through the supply of saving, and ξ_t affects it through the labor supply channel that alters the capital-labor ratio.

4.9 Transition dynamics

This subsection tests prediction **P5** (near-unit-root persistence) by computing the dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ at every point in the historical sample.

The final quantitative exercise examines the stability and persistence properties of the MRCE over the historical sample. These properties are governed by the spectral radius of the matrix $\beta_t \widetilde{\mathcal{G}}_t$, which determines whether the distributional state converges to a unique stationary point or diverges over time. Understanding these dynamics is important for assessing the model's long-run behavior and for interpreting the economic significance of the recovered structural shocks.

Figure 9 plots the dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ over time, along with its two con-

stituent components: the wealth amplification factor $\widetilde{\mathcal{G}}_{11} = 1 + \tilde{r}_t + \tilde{w}_t \xi_t$ (mean 1.111, standard deviation 0.071) and the time-varying discount factor β_t. The dominant eigenvalue has mean 1.027 and standard deviation 0.013, and exceeds unity in 97.4 percent of the sample years—indicating that the U.S. economy spends most of its time in a locally explosive regime. When the eigenvalue is below one, the distributional state converges to a unique stationary point—the economy is in a "stable" regime where shocks are absorbed and the system returns to its long-run configuration. When the eigenvalue exceeds one, the system is locally explosive—the economy is in an "unstable" regime where wealth concentration accelerates and the distributional state diverges from its steady-state value. In practice, the eigenvalue does not remain far above one for extended periods, because the general equilibrium feedback between capital accumulation and returns tends to self-correct: as capital rises, returns fall, which reduces the eigenvalue and restores stability. During the rare stable episodes (eigenvalue < 1), the implied half-life of the distributional state averages 64.8 years, with a range of 23.5 to 106.1 years—underscoring the extreme persistence of distributional dynamics.

The near-unit-root behavior of the dominant eigenvalue has important implications for the propagation of aggregate shocks. When the eigenvalue is close to one, even small perturbations to the distributional state—such as those caused by a TFP shock or a shift in the discount factor—take many periods to dissipate. This generates endogenous persistence in aggregate dynamics that goes beyond the persistence of the underlying structural shocks. In the language of the business cycle literature, the distributional state acts as an internal propagation mechanism that converts transitory shocks into persistent output movements—a role that is analogous to, but distinct from, the capital accumulation mechanism in the standard RBC model.

4.10 Distributional state dynamics

The MRCE's aggregate dynamics are governed by three state variables: aggregate capital K_t and two distributional covariances $\text{cov}_t(z, a)$ and $\text{cov}_t(e, a)$. While the preceding subsections have focused on the aggregate consequences of these covariances, it is informative to examine their evolution directly.

Figure 10 plots the two covariances and their capital-normalized counterparts over the sample. The productivity-wealth covariance $\text{cov}_t(z, a)$ grows from 0.005 in 1948 to 0.343 in 2024, with a mean annual growth rate of 5.5 percent—substantially faster than capital accumulation. This secular increase reflects the compounding nature of productivity sorting: as the economy grows, productive households accumulate wealth at a rate proportional to their existing holdings, generating a self-reinforcing correlation between productivity and wealth. The return-wealth covariance $\text{cov}_t(e, a)$ grows from 0.001 to 0.009, with a mean annual growth rate of 2.8 percent. Both covariances are strongly procyclical: $\text{corr}(\text{cov}(z, a), \log Y) = 0.85$ and $\text{corr}(\text{cov}(e, a), \log Y) = 0.97$, reflecting the fact that wealth accumulation—and hence distributional sorting—accelerates during expansions. Normalizing by aggregate capital, the productivity-sorting ratio $\text{cov}(z, a)/K$ has risen from 0.002 to 0.015 over the sample, while the return-sorting ratio $\text{cov}(e, a)/K$ has remained nearly constant (mean 0.0004, range $[0.0004, 0.0004]$). The constancy of $\text{cov}(e, a)/K$ reflects the near-unit-root dynamics of e, under which the return-wealth covariance scales proportionally with capital.

4.11 TFP growth and inequality dynamics

The preceding subsection documented that the distributional covariances grow faster than capital over the postwar period. This subsection identifies the mechanism: TFP growth is the dominant—and asymmetric—driver of distributional dynamics. The analysis exploits a key advantage of the MRCE: because the distributional state evolves via the closed-form recursion $M_{t+1} = \beta_t \widetilde{\mathcal{G}}_t M_t + [T_t; 0; 0]$, the effect of any

structural shock on inequality can be isolated analytically.

Distributional counterfactuals. I simulate four scenarios forward from 1948, re-evolving the full distributional state $M_t = (K_t, \mathrm{cov}_t(z, a), \mathrm{cov}_t(e, a))$ under each:

- *Baseline*: all three shocks (A_t, β_t, ξ_t) at their historically recovered values.
- *TFP only*: A_t varies historically; β_t and ξ_t are held at their sample means.
- *Fix A*: A_t is held at its 1948 level; β_t and ξ_t vary.
- *No growth*: all shocks fixed at their 1948/mean values (a pure steady-state benchmark).

Unlike the aggregate counterfactuals of the preceding subsection, which hold distributional covariances fixed, these distributional counterfactuals allow the full state to respond endogenously to each scenario.

Figure 11 reports the results. Three findings emerge.

First, TFP is the overwhelming driver of distributional dynamics. Under TFP-only shocks, $\mathrm{cov}(z, a)$ grows by a factor of 71 over the sample—dramatically exceeding the baseline growth factor of 12. The return-wealth covariance $\mathrm{cov}(e, a)$ grows by a factor of 20 under TFP-only, versus 3 in the baseline. In log-difference decomposition, TFP accounts for 228 percent of the total growth in $\mathrm{cov}(z, a)$ and 288 percent of the growth in $\mathrm{cov}(e, a)$—exceeding 100 percent because the other structural shocks (β_t, ξ_t) partially *offset* TFP's inequality-generating effect. This is the paper's central distributional finding: TFP growth is not merely the engine of output growth but is simultaneously—and asymmetrically—the engine of wealth inequality.

Second, without TFP growth the economy collapses distributionally. Under the Fix-A scenario, capital falls to 10 percent of its initial level by 2024, and the productivity-wealth covariance $\mathrm{cov}(z, a)$ shrinks to 4 percent of its 1948 value. The mechanism is straightforward: with A frozen at its 1948 level, wages and returns remain low,

depressing the off-diagonal entries of $\widetilde{\mathcal{G}}$ and shutting down the wealth-sorting channel that sustains distributional heterogeneity.

Third, TFP growth intensifies the rate of wealth sorting *per unit of capital*. The normalized ratio $\text{cov}(z, a)/K$ reaches 0.0105 under TFP-only versus 0.0089 in the baseline. This indicates that TFP growth does not simply scale up all aggregates proportionally; it amplifies the sorting mechanism itself, a form of asymmetry that arises because the off-diagonal entries $\widetilde{\mathcal{G}}_{21} \propto w\xi\sigma_z^2$ and $\widetilde{\mathcal{G}}_{31} \propto r\sigma_e^2$ are increasing in prices—and hence in A.

Analytical decomposition of sorting. The closed-form recursion for $\text{cov}_{t+1}(z, a)$ separates into two channels:

$$\underbrace{\beta_t \widetilde{\mathcal{G}}_{21,t} K_t}_{\text{innovation}} + \underbrace{\beta_t \widetilde{\mathcal{G}}_{22,t} \text{cov}_t(z, a)}_{\text{persistence}}.$$

The innovation term creates new productivity-wealth sorting by channeling aggregate capital into distributional heterogeneity; the persistence term amplifies existing sorting. An analogous decomposition applies to $\text{cov}(e, a)$.

Figure 12 reports the decomposition. Persistence dominates: on average, 85 percent of $\text{cov}(z, a)$ growth comes from the persistence channel and 15 percent from innovation. For $\text{cov}(e, a)$, the persistence share is 99 percent. The right panel of Figure 12 confirms the TFP-inequality nexus: the new sorting rate $\widetilde{\mathcal{G}}_{21}$ is strongly positively correlated with the TFP level A_t, reflecting the mechanism through which higher productivity raises wages and hence amplifies wealth sorting. This provides a structural foundation for the empirical observation that inequality tends to widen during periods of technological progress (Piketty, 2014).

4.12 Distributional impact of TFP growth

A distinctive feature of the MRCE is that the *distribution over trend* is well defined: because the distributional state $(K_t, \mathrm{cov}_t(z,a), \mathrm{cov}_t(e,a))$ evolves via a deterministic 3×3 matrix recursion alongside aggregate capital, one can track, decompose, and compare the evolution of the wealth distribution along the growth path—an analysis that is infeasible in standard heterogeneous-agent models where the distribution is an infinite-dimensional object. This subsection exploits this tractability to make the distributional consequences of TFP growth tangible by computing *who gains how much* from TFP-driven output growth—broken down by wealth percentile.

Steady-state analysis. Table 2 reports the distributional impact of a 1 percent TFP shock at the steady state. For each wealth decile, the table shows the group's share of total wealth, its share of the aggregate wealth gain, the mean percentage wealth gain, and the decomposition into three channels: the capital income channel ($\beta \Delta \tilde{r}\, e_i a_i$), the labor-wealth channel ($\beta \Delta \tilde{w}\, \xi z_i a_i$), and the transfer channel (ΔT, identical for all households). The capital income channel dominates across all deciles, accounting for 64 percent of gains for the bottom decile and 85 percent for the top decile. The transfer channel provides a partially equalizing force—accounting for 17 percent of gains for the bottom decile but less than 1 percent for the top—but is quantitatively insufficient to offset the regressive concentration of capital income gains. The top decile captures 86 percent of the aggregate wealth gain, far exceeding its 82 percent wealth share. The top 1 percent alone captures 71 percent.

Historical period-by-period counterfactual. The steady-state analysis assumes a stationary distribution. To capture the evolving distributional landscape of the U.S. economy, I track 1 million agents through the actual 1948–2024 shock realizations and perform a *period-by-period* counterfactual: at each year t, I compare next-period wealth under the actual TFP level A_t against a counterfactual where TFP remains at

Table 2: Distributional impact of 1% TFP shock at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%) Capital	Labor	Transfer
Bottom 10%	0.8	0.6	0.0829	64.4	18.2	17.3
10–20%	1.1	0.8	0.0869	67.5	19.2	13.2
20–30%	1.2	0.9	0.0892	69.3	19.6	11.1
30–40%	1.4	1.1	0.0911	70.7	19.9	9.5
40–50%	1.6	1.3	0.0930	71.9	20.0	8.1
50–60%	1.9	1.5	0.0950	73.2	20.1	6.7
60–70%	2.3	1.9	0.0972	74.5	20.0	5.5
70–80%	2.9	2.5	0.1000	76.0	19.8	4.1
80–90%	4.4	3.9	0.1037	78.1	19.3	2.6
Top 10%	82.2	85.6	0.1118	84.3	15.5	0.1
Top 5%	78.2	81.9	0.1151	84.5	15.4	0.1
Top 1%	66.4	70.5	0.1205	84.8	15.2	0.0

Notes: Gains computed from a 1% TFP increase at the stationary distribution of 1 million simulated households. "Capital" $= \beta \Delta \tilde{r}\, e_i a_i$; "Labor" $= \beta \Delta \tilde{w}\, \xi z_i a_i$; "Transfer" $= \Delta T$ (identical for all households). Channels reported as percentage of group total gain.

A_{t-1}. Both scenarios use the same distribution and the same (β_t, ξ_t); only the current-year TFP differs. The difference isolates the gain from that year's TFP growth, which is then broken down by wealth decile.

Table 3 reports decade averages of TFP gain shares by wealth decile. Figure 13 visualizes the evolving distribution. The simulation initializes the 1948 wealth distribution to match the U.S. data concentration (Top 10% share $\approx$ 64%, consistent with the Federal Reserve's Distributional Financial Accounts). The central finding is that the top decile captures approximately 70 percent of TFP-driven wealth gains *throughout* the sample period—rising modestly from 70 percent in the 1950s to 73 percent by the 2020s. Within the top decile, however, the composition shifts: the top 1 percent's gain share *falls* from 53 percent (1950s) to 29 percent (2020s), as the 90–99th percentile group captures an increasing share. The bottom 50 percent's gain share declines from 12 percent (1950s) to 5 percent (2020s). Over the full sample, the top decile captured 68 percent and the top 1 percent captured 41 percent of TFP

gains.

The high and stable concentration of gains at the top reflects the multiplicative structure of TFP gains: because the gain from a TFP increase is proportional to $a_i \cdot e_i$ (capital income channel) and $a_i \cdot z_i$ (labor-wealth channel), the top decile—which holds 64–70 percent of wealth and is sorted into high-return, high-productivity households—captures a commensurately disproportionate share of gains. The modest increase from 70 to 73 percent reflects the endogenous further concentration of the wealth distribution over the sample period.

Comparison with U.S. wealth shares. The model's evolving wealth concentration can be compared directly with U.S. data from the Federal Reserve's Distributional Financial Accounts. The simulation is initialized with a 1948 wealth distribution calibrated to match the data concentration (Top 10% share = 64%). Over the 1948–2024 path, the model endogenously generates a further increase in concentration: the top decile's wealth share rises from 64 percent to 70 percent, while the bottom 50 percent's share falls from 16 to 6 percent. In the 2024 data, the top decile holds approximately 64 percent and the bottom 50 percent holds approximately 5 percent of wealth. The model's 2024 top decile share is somewhat above the data (70 vs. 64 percent), while its bottom 50 percent share is close (6 vs. 5 percent). The slight overconcentration at the top reflects the model's near-unit-root return heterogeneity ($\rho_e = 0.999$), which amplifies wealth sorting relative to the data. Despite this, the model captures both the *level* and *stability* of top wealth concentration remarkably well, and the erosion of the bottom 50 percent's share closely matches the data. The key finding is that the MRCE's endogenous wealth-sorting mechanism—driven entirely by the interaction of TFP growth with heterogeneous returns and productivity—generates a degree of concentration that is quantitatively consistent with the U.S. wealth distribution documented by Saez and Zucman (2016) and Piketty (2014).

Table 3: Share of TFP-driven wealth gain by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.6	2.0	2.3	2.6	3.0	3.4	4.0	4.8	6.6	69.7	64.4	52.6
1960s	1.5	1.9	2.3	2.7	3.1	3.5	4.2	5.2	7.2	68.3	62.5	49.7
1970s	1.3	1.8	2.2	2.6	3.1	3.7	4.5	5.7	8.2	66.9	60.2	45.5
1980s	1.1	1.6	2.0	2.4	3.0	3.6	4.6	6.1	9.2	66.4	58.4	41.0
1990s	0.9	1.3	1.7	2.2	2.7	3.4	4.4	6.1	9.8	67.3	58.3	37.8
2000s	0.8	1.2	1.6	2.0	2.5	3.3	4.4	6.2	10.3	67.8	58.0	34.6
2010s	0.6	0.9	1.2	1.6	2.1	2.9	4.0	6.0	10.7	70.0	58.9	30.9
2020s	0.4	0.7	1.0	1.3	1.7	2.4	3.5	5.5	10.6	73.0	61.2	28.6
Full sample	1.1	1.5	1.8	2.2	2.7	3.3	4.2	5.7	9.0	68.4	60.2	41.0

Notes: Each entry reports the average share (in %) of aggregate TFP-driven wealth gain captured by the indicated wealth decile during the specified decade. Period-by-period counterfactual: at each year t, gains are computed as the difference in next-period wealth under A_t versus A_{t-1}. D1 = bottom decile, D10 = top decile.

4.13 Consumption dynamics

The MRCE delivers closed-form aggregate consumption:

$$C_t = \frac{1-\beta_t}{1+\theta}\left[(1+\tilde{r}_t+\tilde{w}_t\xi_t)K_t + \tilde{r}_t \operatorname{cov}_t(e,a) + \tilde{w}_t\xi_t \operatorname{cov}_t(z,a)\right] + \kappa\tilde{w}_t.$$

This expression reveals two properties of the model's consumption dynamics that merit discussion.

Constant marginal propensity to consume. The individual household's MPC out of a one-time transfer is $(1-\beta_t)/(1+\theta)$, which depends on the discount factor and the leisure weight but not on the household's wealth level. At the steady-state calibration ($\beta = 0.96$, $\theta = 2$), the annual MPC is 1.3 percent. The cumulative MPC reaches 6.6 percent after five years and 26.3 percent after twenty years, reflecting the near-unit-root wealth dynamics that cause transfers to be consumed only gradually. The constant MPC is a direct consequence of the linear policy functions that enable the MRCE's analytical tractability: because the saving rate is independent of wealth, all households—regardless of their position in the wealth distribution—respond iden-

tically to a marginal transfer. This stands in contrast to models with concave consumption functions, where borrowing-constrained "hand-to-mouth" households exhibit MPCs of 0.3–0.5 (Kaplan et al., 2018). The model's low, uniform MPC implies that fiscal transfers are a less powerful stabilization tool than in models with binding borrowing constraints—a limitation that is the price of analytical tractability.

Procyclical MPC. The time-varying MPC $(1-\beta_t)/(1+\theta)$ inherits the procyclicality of $(1-\beta_t)$: since β_t is countercyclical (prediction **P4**), the MPC is procyclical, averaging 2.7 percent during expansions versus 1.2 percent during recessions. This pattern—households consume a larger fraction of marginal income during booms and a smaller fraction during downturns—provides an additional stabilizing mechanism, as it dampens the amplification of demand shocks.

4.14 Historical episode analysis

The exact inversion provides a structural narrative for each recession in the sample: the recovered series $\{A_t, \beta_t, \xi_t\}$ reveal which structural forces drove each downturn. Figure 14 displays the behavior of TFP and the discount factor around three key recessions that illustrate distinct macroeconomic mechanisms.

1973–75 (Oil Crisis). Output declined by 0.75 percent from peak to trough while TFP continued to grow (at 1.2 percent per year, below the sample average of 2.1 percent). The discount factor was essentially unchanged ($\Delta\beta = -0.005$). This pattern is consistent with a supply-side recession: a slowdown in productivity growth reduced the economy's trend growth rate, while demand-side forces played a minimal role. The structural narrative aligns with the conventional interpretation of the 1970s recession as driven by oil price shocks that reduced productive efficiency.

2007–09 (Great Recession). Output declined by 2.5 percent from peak to trough—the largest recession in the sample. TFP continued to grow at its trend rate (2.0 per-

cent per year), suggesting that the recession was not driven by a collapse in productive capacity. Instead, the discount factor exhibited an extraordinary pattern: β_t peaked at 0.993 in 2007 and then dropped sharply to 0.834 by 2009—a swing of 0.159. The pre-crisis peak in β reflects a period of historically high saving and capital accumulation (consistent with the global savings glut narrative), while the subsequent collapse reflects a dramatic shift toward present consumption—potentially driven by the financial crisis, the housing wealth shock, and heightened uncertainty. The concurrent jump in ξ_t from near zero to its upper bound (0.026) suggests that the wealth-leisure channel was activated as wealthy households adjusted their labor supply in response to large wealth losses.

2020 (COVID-19). Output declined by 2.1 percent in a single year. TFP grew above its trend rate (2.6 percent), ruling out a supply-side explanation at the aggregate level. The discount factor was already low ($\beta = 0.806$) and changed minimally ($\Delta\beta = -0.006$). This suggests that the COVID recession, in the model's interpretation, was not driven by a shift in patience or saving behavior but rather by an exogenous disruption that the model's three structural shocks do not fully capture—consistent with the view that the pandemic was a sui generis shock that operated primarily through sectoral reallocation and public health constraints rather than through the standard macroeconomic channels.

4.15 Robustness

Table 4 reports the sensitivity of the model's key steady-state properties to alternative calibrations. Three findings stand out. First, the approximate aggregation result (prediction **P1**) is extremely robust: I^{rep} accounts for 93–99 percent of total investment across all parameter configurations examined. Second, the near-unit-root persistence of the distributional state (prediction **P5**) is robust to all parameters except the transfer share η: reducing η to 0.005 (less redistribution) increases the

half-life to 1,480 years, while raising η to 0.05 (more redistribution) reduces it to 157 years. This sensitivity is economically intuitive: redistribution pulls wealth toward the mean, accelerating convergence, while less redistribution allows wealth sorting to persist longer. Third, the leisure weight θ and productivity persistence ρ_z have essentially no effect on any of the key metrics, confirming that the model's aggregate predictions are driven by the capital accumulation mechanism (α, β, δ) and the redistributive environment (η), not by the labor supply elasticity or the details of the idiosyncratic process.

Summary of quantitative findings

Table 5 summarizes the quantitative evaluation of the five testable predictions.

5 Extensions

The baseline model uses Stone-Geary preferences with log utility. A natural question is whether the main results—the existence of the MRCE, belief invariance, and the closed-form moment recursion—are artifacts of the log specification, or whether they extend to richer preference structures that are commonly used in the quantitative macro and finance literatures. This section discusses two important alternative preference specifications: CRRA-GHH utility, which is widely used in the business cycle literature for its ability to eliminate wealth effects on labor supply, and Epstein-Zin recursive utility, which is the workhorse specification in the asset pricing literature for its ability to separate risk aversion from the elasticity of intertemporal substitution.

5.1 CRRA-GHH utility with $e = 1$

When return heterogeneity is shut down ($e = 1$) and households have GHH (Greenwood-Hercowitz-Huffman) utility with CRRA curvature σ, the MRCE continues to exist, but the belief-invariance property does not hold. The reason is economically intuitive:

Table 4: Sensitivity of steady-state properties to alternative calibrations

Parameter	Value	K/Y	r (%)	$\lambda_{\max}$	I^{rep} (%)	Half-life (yr)
Baseline		3.23	3.22	0.9988	98.2	563
α	0.3	2.89	3.39	0.9986	98.0	503
	0.36	3.61	2.98	0.9989	98.4	629
	0.4	4.23	2.46	0.9991	98.6	736
σ_e	0.001	3.23	3.22	0.9988	98.2	561
	0.005	3.23	3.22	0.9988	98.2	561
	0.02	3.23	3.22	0.9988	98.1	569
	0.05	3.24	3.18	0.9989	97.8	605
	0.1	3.27	3.08	0.9990	97.0	686
θ	1	3.23	3.22	0.9988	98.2	563
	1.5	3.23	3.22	0.9988	98.2	563
	3	3.23	3.22	0.9988	98.2	563
	5	3.23	3.22	0.9988	98.2	563
η	0.005	3.19	3.35	0.9995	99.3	1480
	0.01	3.21	3.27	0.9991	98.6	745
	0.02	3.26	3.12	0.9982	97.3	378
	0.03	3.31	2.97	0.9973	96.1	255
	0.05	3.41	2.68	0.9956	93.7	157
β	0.94	2.42	6.65	0.9984	97.6	425
	0.95	2.76	4.94	0.9986	97.9	484
	0.97	3.89	1.47	0.9990	98.5	677
	0.98	4.96	-0.35	0.9992	98.8	862
ρ_z	0.9	3.23	3.22	0.9988	98.2	562
	0.93	3.23	3.22	0.9988	98.2	563
	0.97	3.23	3.22	0.9988	98.2	563
	0.99	3.23	3.20	0.9988	98.0	566

Notes: Each row varies one parameter from the baseline calibration. K/Y: capital-output ratio. r: real interest rate. $\lambda_{\max}$: dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$. I^{rep}: representative-agent share of total investment. Half-life: years for the distributional state to halve its deviation from steady state.

away from log utility ($\sigma \neq 1$), the income and substitution effects of changes in future returns no longer cancel. When $\sigma > 1$, the income effect dominates, and households save *more* when they expect high future returns (because the wealth effect of higher future income reduces the need for current saving by less than the substitution effect

Table 5: Quantitative evaluation of testable predictions (U.S., 1948–2024)

	Prediction & test	Result
P1	*Approximate aggregation.*	
	I^{rep} share of mean investment	98.4%
	I^{rep} share of investment variance	93.3%
P2	*State-dependent amplification.*	
	Output impact, 2022 vs. 1952 state	+37%
	Cumulative 10-yr loss, 2022 vs. 1952	+67%
P3	*Return risk premium.*	
	Premium at baseline $\sigma_e = 0.01$	−0.001 bp
	σ_e threshold for 1 bp premium	≈ 0.83
P4	*Countercyclical patience.*	
	$\text{corr}(\beta_t, \log Y_t)$	−0.72
	Mean β_t: recessions vs. expansions	0.965 vs. 0.918
P5	*Near-unit-root persistence.*	
	Mean dominant eigenvalue	1.027
	Half-life (stable episodes)	64.8 years

increases it). When $\sigma < 1$, the substitution effect dominates, and households save *less* when they expect high future returns. In either case, the saving rate depends on the expected future return, which means that household beliefs about aggregate dynamics feed into current decisions. The MRCE can still be solved as a fixed-point problem in the moments—because the policy functions remain linear in wealth, ensuring that the moment structure is closed—but the simplicity of the belief-invariant characterization is lost. Computing the equilibrium requires specifying the stochastic process for aggregate shocks and solving for the expectation-dependent saving rate, much as in a standard incomplete-markets model.

The GHH specification is nonetheless valuable because it eliminates the wealth effect on labor supply. In the baseline Stone-Geary model, wealthy households work less because they value leisure more highly (through the $\kappa + \xi a$ term in the time endowment). With GHH preferences, labor supply depends only on the current wage,

not on wealth, which simplifies the aggregate labor equation and may be appropriate for applications where the wealth effect on hours is not a first-order concern.

5.2 Recursive (Epstein-Zin) utility

A more productive extension—and one that preserves the full analytical tractability of the baseline model—is to Epstein-Zin preferences with unit elasticity of intertemporal substitution (EIS = 1). This is arguably the most important extension, because it allows the model to address one of the central puzzles in asset pricing: the equity premium. Under expected utility with log preferences, the coefficient of relative risk aversion is fixed at one, which generates an equity premium that is far too small relative to historical data. Epstein-Zin preferences break this tight link by introducing a separate risk aversion parameter γ that can be set independently of the EIS. Under these preferences, the flow composite is:

$$\tilde{c} \equiv c - \kappa z w(X)(1-\tau), \qquad m \equiv (\kappa + \xi a) - n, \qquad X^{\text{comp}} \equiv \tilde{c}\, m^{\theta}.$$

The intratemporal optimality condition implies $m = (\theta/(w(X)(1-\tau)z))\tilde{c}$, so:

$$X^{\text{comp}} = \tilde{c}^{\,1+\theta} \left(\frac{\theta}{w(X)(1-\tau)z} \right)^{\theta}.$$

The recursive value function satisfies:

$$\log V_t = (1-\beta)\log X_t + \beta \log \left(\mathbb{E}_t \left[V_{t+1}^{1-\gamma} \right] \right)^{\frac{1}{1-\gamma}}, \qquad \beta \in (0,1), \quad \gamma > 0.$$

The critical result is that, with EIS = 1, the optimal saving share $s_a(s)$ satisfies the same type of fixed-point condition as in the log case:

$$s_a(s) = \left[1 + \left(\frac{\beta(1+\theta)^p\, \bar{B}(s)}{C(s)} \right)^{\frac{1}{p-1}} \right]^{-1},$$

where $p \equiv (1+\theta)(1-\sigma)$, $C(s) \equiv (\theta/(w(X)(1-\tau)z))^{\theta(1-\sigma)}$, and $\bar{B}(s) \equiv \mathbb{E}[B(s') \mid s]$ is a conditional expectation that can be computed from the fixed-point system:

$$B(s) = \Pi(s)^p \left[C(s) \left(\frac{1 - s_a(s)}{1+\theta} \right)^p + \beta\, s_a(s)^p\, \bar{B}(s) \right],$$

with $\Pi(s) \equiv (1 + r(X)(1-\tau))e + w(X)(1-\tau)z\xi$.

The key takeaway is that the MRCE and BRCE continue to hold under Epstein-Zin preferences with unit EIS. The saving share is still independent of the household's wealth level (though it may depend on the exogenous state $s = (z, e, A)$), and the law of motion for the aggregate moments retains its closed-form matrix structure. The parameter γ controls the household's aversion to risk over consumption *lotteries* (a cross-sectional concept), while β and θ control the intertemporal and intratemporal tradeoffs (time-series concepts). Separating these roles—which is precisely what Epstein-Zin preferences achieve—allows the model to match both the equity premium (through γ) and the saving rate (through β) without forcing them to be determined by a single parameter.

This extension is particularly valuable in light of the asset pricing implications discussed in Section 3.9. With log utility ($\gamma = 1$), the model generates a risk premium from idiosyncratic return heterogeneity, but its magnitude is limited by the low degree of risk aversion. With Epstein-Zin preferences and $\gamma \gg 1$, the same wealth-sorting mechanism generates a substantially larger risk premium, because households are more averse to the consumption risk induced by return heterogeneity. The combination of the MRCE's analytical moment recursion with Epstein-Zin preferences thus provides a tractable framework for studying the joint determination of asset prices, wealth inequality, and aggregate dynamics—a problem that has traditionally required large-scale numerical methods.

6 Conclusion

This paper has developed an analytical framework for studying aggregate fluctuations in an economy with heterogeneous households. The central contribution is the moment-recursive competitive equilibrium (MRCE), in which the entire cross-sectional distribution influences aggregate outcomes through exactly three summary statistics: aggregate capital and two covariances measuring the alignment between wealth and idiosyncratic characteristics. These moments follow a closed 3×3 matrix recursion that can be solved without simulation, approximation, or numerical iteration.

The framework delivers three sets of theoretical insights.

First, distributional covariances generate state-dependent aggregate dynamics. The response of the economy to a productivity shock depends not only on the current level of capital but also on the current alignment between wealth and idiosyncratic characteristics. When wealth is concentrated among high-return or high-productivity households, the same aggregate shock produces a different composition of consumption and investment than when wealth is more evenly distributed. This state dependence is an exact analytical result, not a numerical observation, and it provides a direct counterpart to the approximate aggregation finding of Krusell and Smith (1998)—extending it to a richer setting with return heterogeneity and endogenous labor supply, and establishing it as a theorem rather than a simulation result.

Second, idiosyncratic return heterogeneity, which is traditionally treated as diversifiable, carries a positive risk premium because it affects aggregate outcomes through the wealth-sorting channel. Households with persistently high returns accumulate more wealth, which alters the composition of the economy and feeds back into aggregate factor prices. This mechanism provides a new source of systematic risk that does not require aggregate shocks to idiosyncratic volatility—the cross-sectional dispersion of returns is sufficient to generate aggregate consequences through endogenous wealth sorting.

Third, the equilibrium is invariant to household beliefs about aggregate dynamics. This belief-invariance property means that growth and business cycle analysis can be conducted within a single framework without specifying how households form expectations about the aggregate economy. The same structural equations govern the equilibrium regardless of whether the economy is on a balanced growth path, experiencing a transitory recession, or undergoing a structural transformation. This property is particularly valuable for the quantitative application, where it eliminates the need to specify an expectation formation mechanism when mapping from data to structural shocks.

The closed-form structure enables a distinctive quantitative application: an exact general equilibrium inversion that maps observed macroeconomic data—output, capital, and hours—to the structural shocks that generated them. Applied to U.S. data over 1948–2024, the inversion recovers three structural series. Aggregate TFP follows well-known historical narratives, growing steadily during the postwar expansion, slowing during the 1970s productivity slowdown, accelerating during the IT revolution, and stagnating after the Great Recession. The time-varying discount factor is countercyclical, rising during recessions and falling during expansions, consistent with demand-side interpretations of business cycle fluctuations. Counterfactual experiments reveal that TFP is the dominant driver of trend growth and wage dynamics, while the discount factor is the primary source of cyclical variation in output and employment. The investment decomposition shows that the representative-agent channel accounts for the bulk of aggregate capital formation, while the return-sorting and productivity-sorting channels provide nonnegligible distributional corrections.

Several directions for future research are natural. First, the model could be embedded in a New Keynesian setting with nominal rigidities to study the interaction between monetary policy and distributional dynamics. The MRCE's analytical characterization of how the wealth distribution shapes aggregate consumption and labor supply could shed new light on the transmission mechanisms of monetary policy in

economies with substantial household heterogeneity—a question at the frontier of the HANK literature. Second, the closed-form dependence of aggregate outcomes on the tax rate τ and the transfer share η could be exploited to design optimal tax and transfer policies that account for the general equilibrium feedback between redistribution and aggregate capital accumulation. The model's analytical structure makes it possible to compute the welfare effects of policy changes without simulation, which could significantly lower the barrier to rigorous distributional analysis in policy settings. Third, the exact inversion methodology could be extended to panel data across countries, taking advantage of the fact that the model's sufficient statistics—aggregate capital, output, and hours—are available in international databases such as the Penn World Tables. Cross-country variation in the recovered structural shocks could illuminate the sources of the large differences in income levels and growth rates that characterize the global economy. Fourth, the Epstein-Zin extension opens the door to asset pricing applications in which the risk premium, the wealth distribution, and aggregate dynamics are jointly determined in a single analytical framework. I leave these extensions for future work.

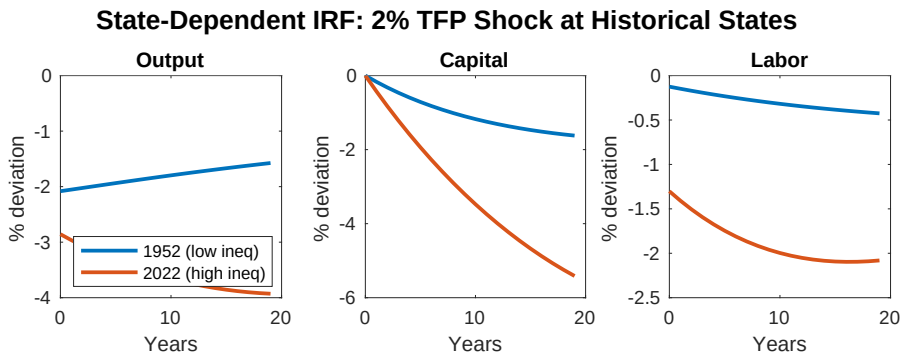

Figure 5: State-dependent impulse responses to a 2% negative TFP shock. Blue: launched from the 1952 distributional state (low inequality). Red: launched from the 2022 distributional state (high inequality). The same shock produces substantially larger and more persistent output losses when wealth inequality is high.

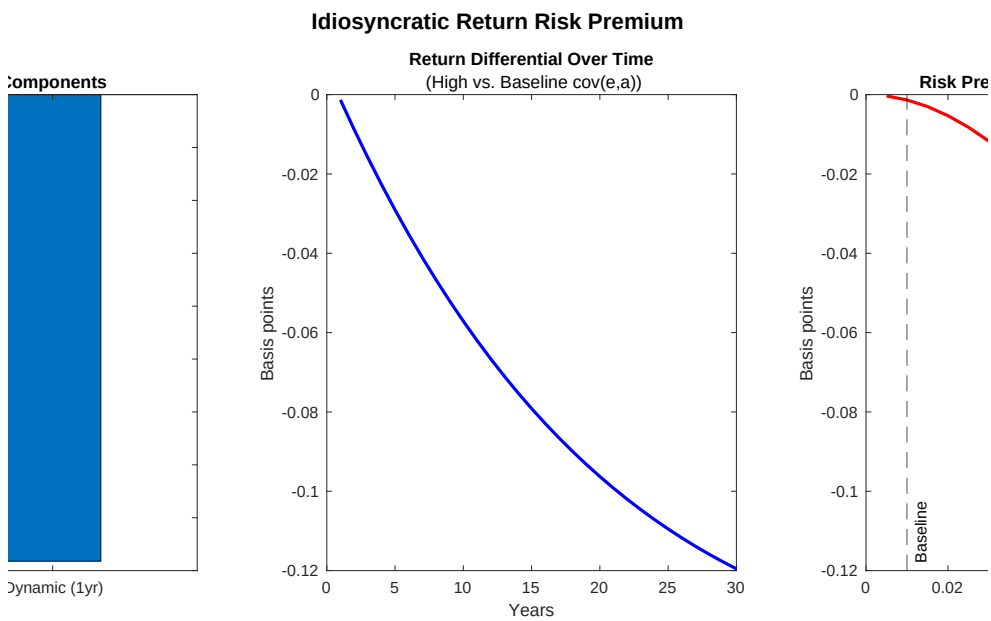

Figure 6: Return risk premium sensitivity. Left: static vs. dynamic premium components at steady state. Center: cumulative return differential between a high-cov(e, a) economy and the baseline over 30 years. Right: premium as a function of return dispersion σ_e. The threshold for 1 bp is $\sigma_e \approx 0.83$; the economy becomes unstable before reaching 10 bp. All values in basis points.

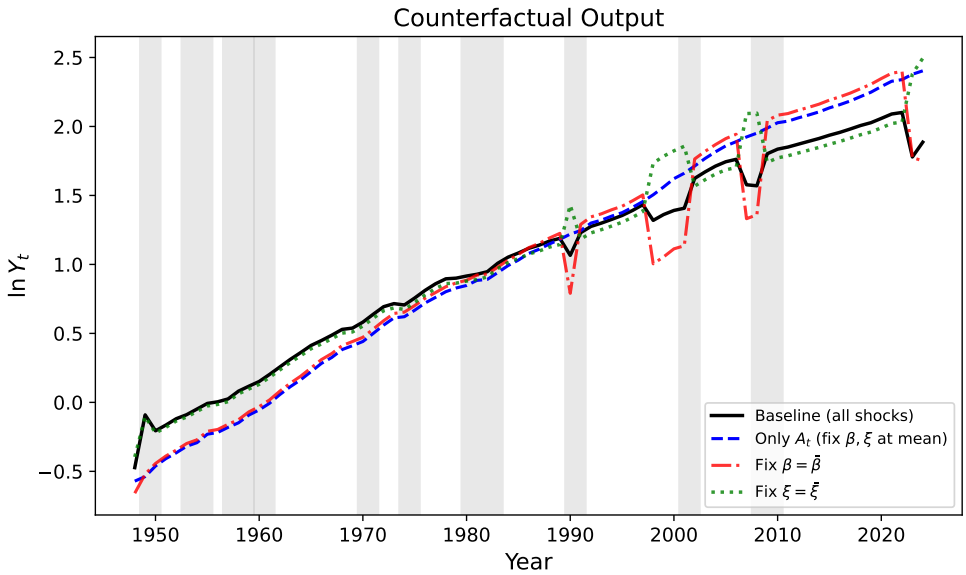

Figure 7: Counterfactual output paths. Baseline: all structural shocks at historical values. TFP only: β_t and ξ_t fixed at sample means. Fix β: β_t fixed, A_t and ξ_t vary. Fix ξ: ξ_t fixed, A_t and β_t vary. Shaded areas indicate NBER recession dates.

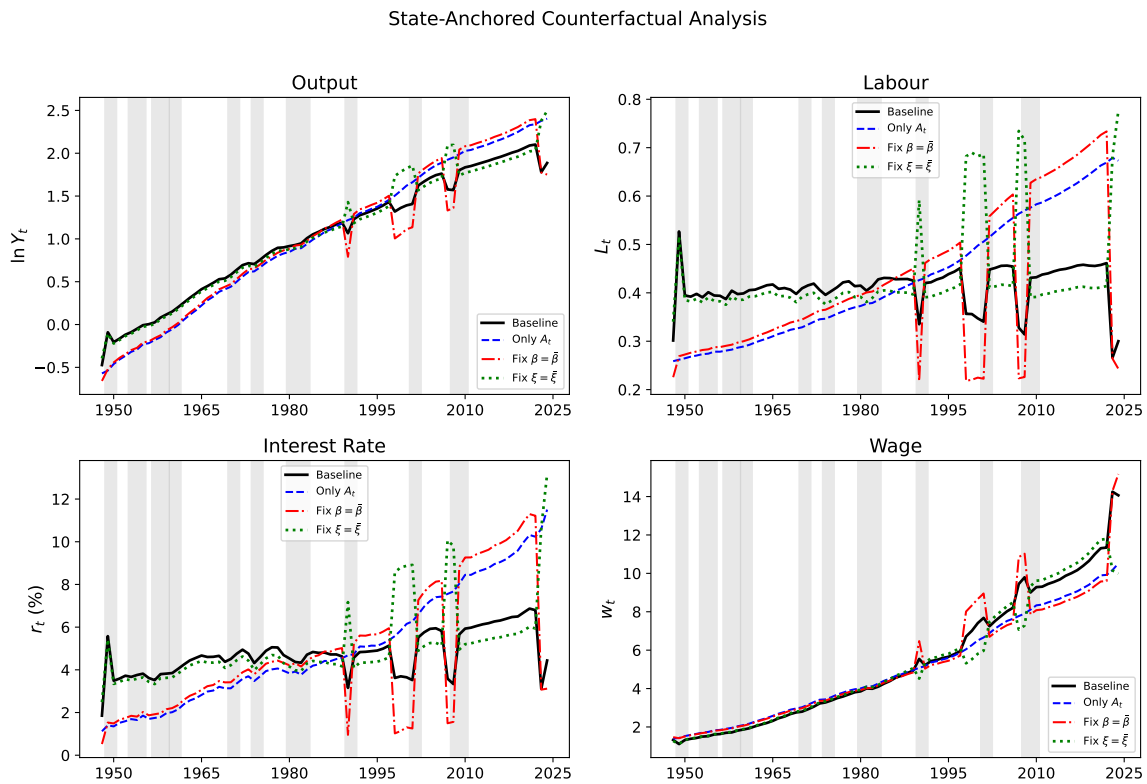

Figure 8: Counterfactual comparison across four macroeconomic variables: output (log), labor supply, interest rate, and wage. Each panel compares the baseline with the three counterfactual scenarios. Shaded areas indicate NBER recession dates.

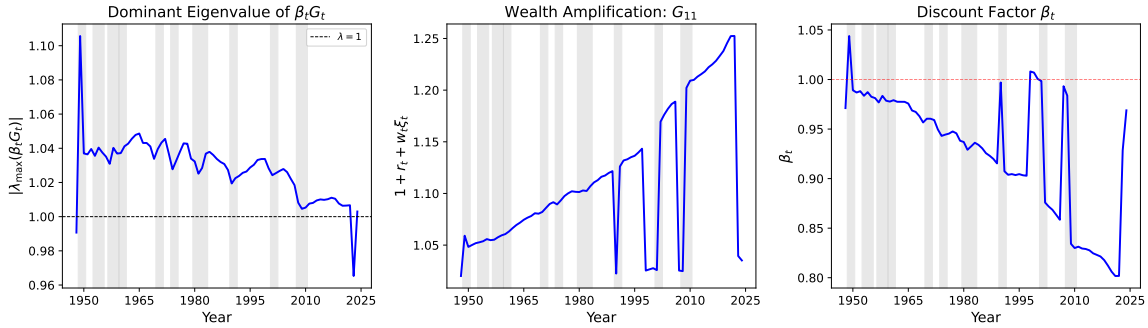

Figure 9: Transition dynamics of the MRCE. Top: dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$. Middle: wealth amplification factor $\widetilde{\mathcal{G}}_{11} = 1 + \tilde{r}_t + \tilde{w}_t \xi_t$. Bottom: time-varying discount factor β_t. Shaded areas indicate NBER recession dates.

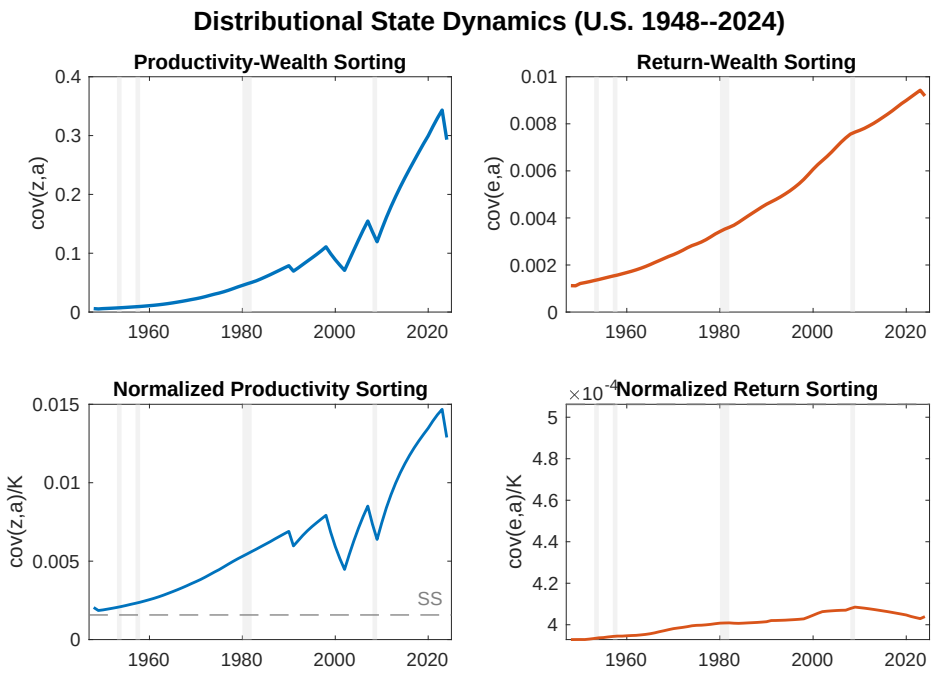

Figure 10: Distributional state dynamics, 1948–2024. Top row: raw covariances $\mathrm{cov}(z, a)$ (left) and $\mathrm{cov}(e, a)$ (right). Bottom row: capital-normalized ratios $\mathrm{cov}(z, a)/K$ and $\mathrm{cov}(e, a)/K$, with steady-state values shown for reference. Shaded areas indicate NBER recession dates.

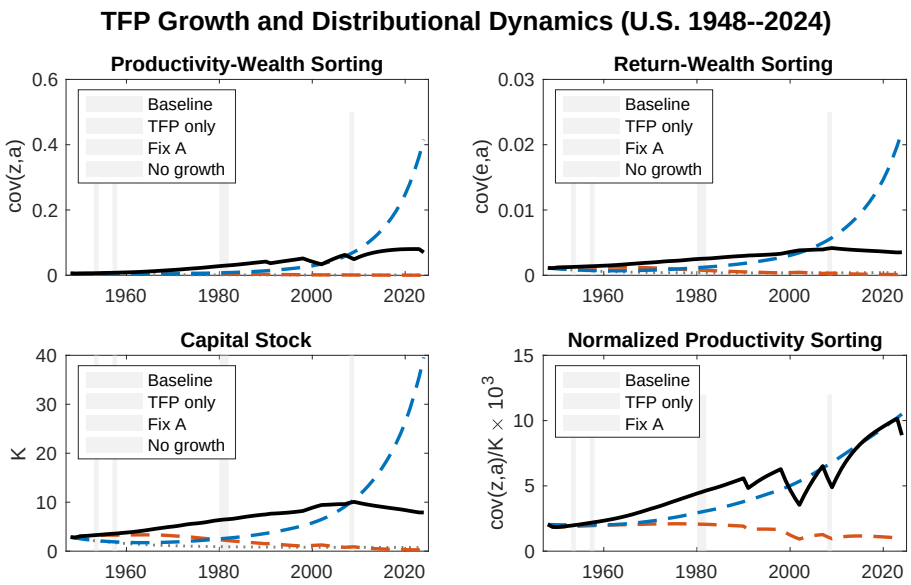

Figure 11: TFP growth and distributional dynamics. Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital K. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. TFP-only shocks generate distributional dynamics that far exceed the baseline, while removing TFP collapses both capital and distributional heterogeneity. Shaded areas indicate NBER recessions.

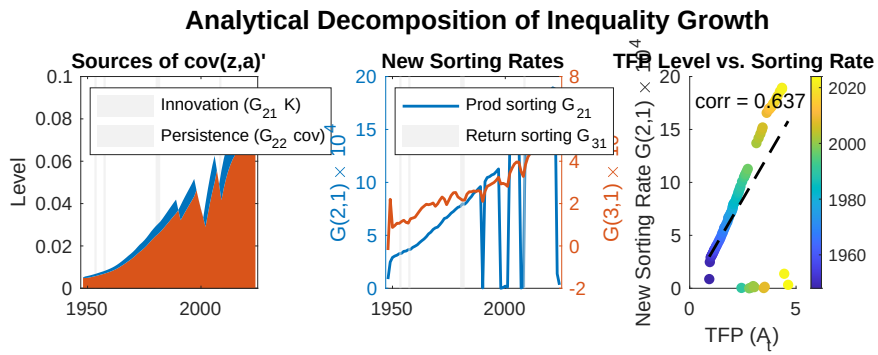

Figure 12: Analytical decomposition of distributional sorting. Left: contributions of innovation ($\widetilde{\mathcal{G}}_{21}K$) and persistence ($\widetilde{\mathcal{G}}_{22}\text{cov}(z,a)$) to the evolution of the productivity-wealth covariance. Center: time paths of the new sorting rates $\widetilde{\mathcal{G}}_{21}$ (productivity) and $\widetilde{\mathcal{G}}_{31}$ (return). Right: scatter plot of TFP level vs. new sorting rate, showing the positive relationship that drives the TFP–inequality nexus.

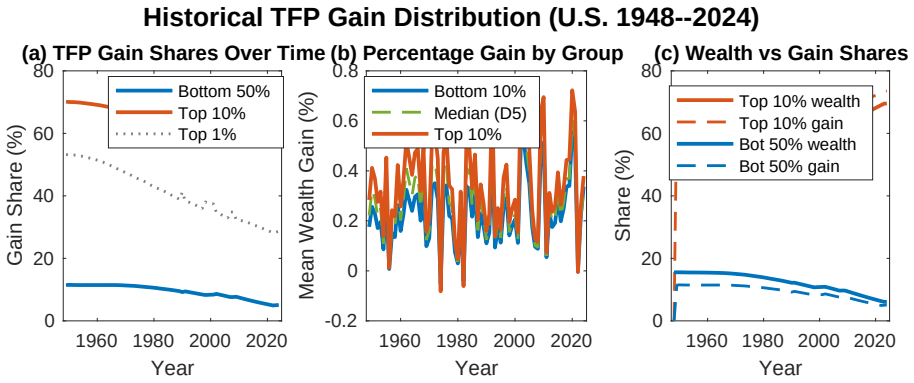

Figure 13: Historical distribution of TFP gains across wealth percentiles, U.S. 1948–2024. Left: gain share of bottom 50%, top 10%, and top 1% over time. Center: mean percentage wealth gain for bottom 10%, median (D5), and top 10%. Right: wealth share versus gain share for top 10% and bottom 50%.

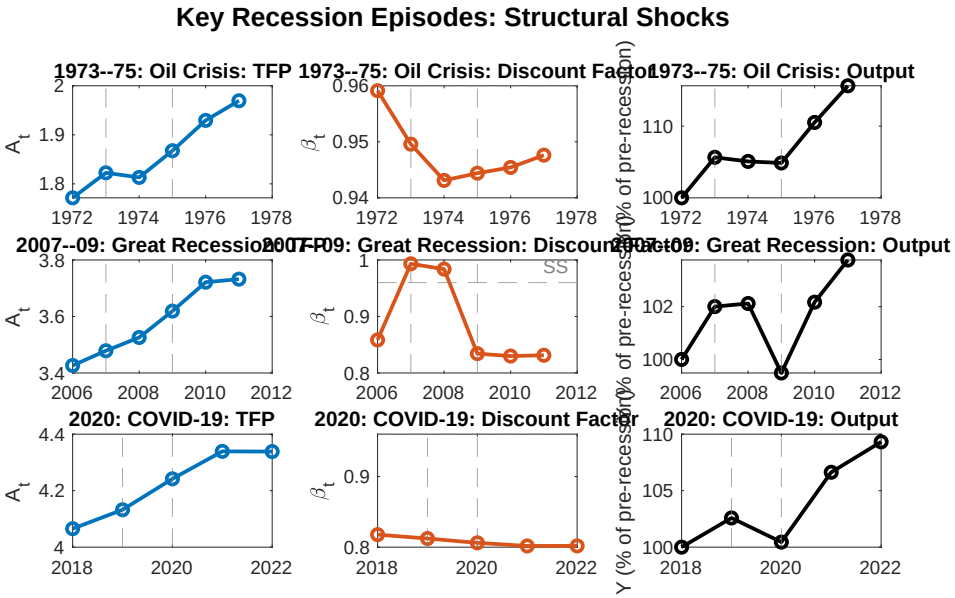

Figure 14: Structural shocks around three key recessions. Each row shows one recession episode: 1973–75 (Oil Crisis), 2007–09 (Great Recession), and 2020 (COVID-19). Left: TFP. Center: discount factor β_t, with steady-state value for reference. Right: output indexed to pre-recession level. Dashed lines mark peak and trough years.

References

Aguiar, M. and E. Hurst (2007). Measuring trends in leisure: The allocation of time over five decades. *Quarterly Journal of Economics 122*(3), 969–1006.

Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics 109*(3), 659–684.

Angeletos, G.-M. and J. La'O (2013). Sentiments. *Econometrica 81*(2), 739–779.

Auclert, A., B. Bardóczy, M. Rognlie, and L. Straub (2021). Using the sequence-space Jacobian to solve and estimate heterogeneous-agent models. *Econometrica 89*(5), 2375–2408.

Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? Risk, return, and skill in household wealth. *American Economic Review 110*(9), 2703–2747.

Bewley, T. (1986). Stationary monetary equilibrium with a continuum of independently fluctuating consumers. In W. Hildenbrand and A. Mas-Colell (Eds.), *Contributions to Mathematical Economics in Honor of Gérard Debreu*, pp. 79–102. North-Holland.

Bilbiie, F. O. (2020). The new Keynesian cross. *Journal of Monetary Economics 114*, 90–108.

Bloom, N., M. Floetotto, N. Jaimovich, I. Saporta-Eksten, and S. J. Terry (2018). Really Uncertain Business Cycles. *Econometrica 86*(3), 1031–1065.

Brock, W. A. and L. J. Mirman (1972). Optimal economic growth and uncertainty: The discounted case. *Journal of Economic Theory 4*(3), 479–513.

Chari, V. V., P. J. Kehoe, and E. R. McGrattan (2007). Business cycle accounting. *Econometrica 75*(3), 781–836.

Constantinides, G. M. and D. Duffie (1996). Asset pricing with heterogeneous consumers. *Journal of Political Economy 104*(2), 219–240.

Cooley, T. F. and E. C. Prescott (1995). Economic growth and business cycles. In T. F. Cooley (Ed.), *Frontiers of Business Cycle Research*, pp. 1–38. Princeton University Press.

Dynan, K. E., J. Skinner, and S. P. Zeldes (2004). Do the rich save more? *Journal of Political Economy 112*(2), 397–444.

Eggertsson, G. B. and P. Krugman (2012). Debt, deleveraging, and the liquidity trap: A Fisher–Minsky–Koo approach. *Quarterly Journal of Economics 127*(3), 1469–1513.

Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica 88*(1), 115–170.

Farhi, E. and I. Werning (2019). Monetary policy, bounded rationality, and incomplete markets. *American Economic Review 109*(11), 3887–3928.

Guvenen, F. (2009). An empirical investigation of labor income processes. *Review of Economic Dynamics 12*(1), 58–79.

Huggett, M. (1993). The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control 17*(5–6), 953–969.

Kaplan, G., B. Moll, and G. L. Violante (2018). Monetary policy according to HANK. *American Economic Review 108*(3), 697–743.

Kesten, H. (1973). Random difference equations and renewal theory for products of random matrices. *Acta Mathematica 131*, 207–248.

Kuhn, M., M. Schularick, and U. I. Steins (2020). Income and wealth inequality in America, 1949–2016. *Journal of Political Economy 128*(9), 3469–3519.

Menzio, G. and S. Shi (2010). Block recursive equilibria for stochastic models of search on the job. *Journal of Economic Theory 145*(4), 1453–1494.

Piketty, T. (2014). *Capital in the Twenty-First Century*. Cambridge, MA: Harvard University Press.

Prescott, E. C. (1986). Theory ahead of business-cycle measurement. *Carnegie-Rochester Conference Series on Public Policy 25*, 11–44.

Ravn, M. O. and V. Sterk (2021). Macroeconomic fluctuations with HANK and SAM: An analytical approach. *Journal of the European Economic Association 19*(2), 1162–1202.

Saez, E. and G. Zucman (2016). Wealth inequality in the United States since 1913: Evidence from capitalized income tax data. *Quarterly Journal of Economics 131*(2), 519–578.

Schmidt, L. D. W. (2016). Climbing and falling off the ladder: Asset pricing implications of labor market event risk. *Working Paper*.

Storesletten, K., C. I. Telmer, and A. Yaron (2004). Cyclical dynamics in idiosyncratic labor market risk. *Journal of Political Economy 112*(3), 695–717.

Summers, L. H. (2014). U.s. economic prospects: Secular stagnation, hysteresis, and the zero lower bound. *Business Economics 49*(2), 65–73.